

# Universe24: an integer cellular automaton read through clicks, the formulas it shares with nature and the ones it adds

Alon Gonen 

Independent researcher, Haifa, Israel.

Corresponding author(s). E-mail(s): [alon@defounder.ai](mailto:alon@defounder.ai);

## Abstract

Universe24 is one line of integer arithmetic: a cubic lattice of Nodes, two integer levels per field, stepped by one recurrence, Rule3, from a Node's present, past and six neighbours, reversibly and covariantly under the cube's 48. The lattice is read only through declared counting regions, NodeDetectors, its clicks the only measurements, each the meeting of the forward field with the transition read backward. From it follow, in lattice form, the Klein-Gordon equation with Schrödinger's slow limit, Planck's relation in the band's form and, under one named assumption fixed by two measurements, the exponential weak-field metric,  $\gamma = \beta = 1$ , and, with no number put in, frame dragging's form,  $\alpha_1 = \alpha_2 = 0$  and  $\eta_N = 0$  at nature's gap. A field's rational pair  $[\mathbf{num}, \mathbf{den}]$ ,  $\cos \omega_0 = \mathbf{num}/\mathbf{den}$ , fixes its inertial mass  $m^* = 3 \tan \omega_0$  and the post-Newtonian parameters  $\gamma_K = \mathbf{den}/\mathbf{num}$  and  $\beta_K$ , closing to 1 with the gap. The lattice's own terms are predictions under their conditions: a black-hole shadow radius  $2e m$ , light's gravitational source at  $E/c^2$ , the metric's second order, light undragged by a rotating mass. Bell's and Greenberger-Horne-Zeilinger correlations follow from a declared counting rule. Its misses are named, a decelerating cosmology among them. Universe24 is one world of a method that can build others; the paper claims that it helps compute things in nature, not that nature is such a lattice.

**Keywords:** integer cellular automaton, reversible lattice dynamics, quantum measurement, Bell inequalities, weak-field gravity on a lattice

# 1 Introduction

## 1.1 The question and the claim

This paper is about one method of computing physics. A cubic lattice of Nodes is stepped by one local rule in whole numbers, Rule3, which is exactly reversible, and it is read only through clicks, the reports of declared counting regions that the model calls NodeDetectors. In the words of the title: the integer cellular automaton is the lattice stepped by Rule3; Universe24 is its clicks, written by the NodeDetectors from the fields that its configuration files declare; the formulas it shares with nature are the known formulas reached from the one line in lattice form, the Klein-Gordon equation with Schrödinger's slow limit, Planck's relation in the band's form (the adiabatic invariant [1]) and, under one named assumption, the exponential weak-field metric of gravity (Section 5.1); and the ones it adds are the lattice's own terms, which depart from the continuum or have no counterpart there (Section 5.2). The paper has one claim: that Universe24 helps compute things in nature. Its content is what the model computes of nature set against measurement, each quantity with its status and its number, the hits and the misses alike; the method is how it computes and not the claim. The paper does not claim that nature is such a lattice.

The criterion is the same throughout: a derived, click-fenced formula that disagrees with a measurement beyond its stated order refutes the model at that formula, and Section 8 lists the refutations found. Universe24 is one world built by the paper's method: one declared universe file, one run read by its NodeDetectors (the two slits, Section 6), and the forms of Sections 4 and 5 derived from the law's lines and computed from no run. Other worlds closer to nature can be built by the same method, and building one needs the calculations named in Section 8.3, which this work does not do.

The paper starts from one statement and never contradicts it (assumption, the paper's first statement; clicks; the key of Section 1.3): Universe24 is the meeting of the past with the future; a click is that meeting, the one measurement there is in the model; and in it only clicks are seen. A Node holds a field's present and its past, two integer levels, and the rule makes the field's future from these two and from the six neighbouring Nodes. Run forward from the source Node of its initial data, the split events are the future, every path at once; read backward from a click that happened, by the same rule, they are the past, and through two slits both of them pass through both gaps. A click is where the two meet at a declared NodeDetector, and which meeting happens is the one thing the model does not determine, the one non-local element of the description. In the model, what looks like a wave is the future; what looks like a particle is the present, the click; the track is the past.

In the model, whoever sees is itself a cluster of bound states on the lattice, a Node reads its six Ports and nothing else, and beyond the cluster's tail nothing reaches it except what a NodeDetector reports, one quantum credited at a meeting; what stands on the lattice between clicks is not a fact of Universe24 but its possibilities, and a fact is a realised event, a quantum credited at a click. So the paper has three words and keeps them apart. The lattice is the lattice stepped by Rule3; it computes. The NodeDetector is the gate between the lattice and Universe24: it reads the lattice and writes the clicks that happened. Universe24 is those clicks, the world of the model. A

lattice reading, a level, a support, a count at a Node, is a diagnostic of the computation and is labelled so, and only a click pattern is compared with a measured pattern of nature. A deterministic, reversible integer lattice read only through clicks gives the statistics of one quantum computed here, one quantum per click, the two slits' pattern and Bell's values, from a wave on the lattice and a declared counting rule; Born's rule is put in, and the Hilbert space beyond one quantum, spin and the many-body atom are not reached (derived for the wave and interference, exclusivity derived under the credit's one draw, the NodeDetector's declaration; an assumption, fitted, for the counting rule, the draw and the conditional; the misses by name, Section 8); the backward reading fixes no number that the forward counting rule does not, the joint energy being computed from the forward configurations alone (Section 2.5).

## 1.2 What the paper adds, and its precursors

Three things, each with its status in its sentence and in the claims list of the supplementary material: the method, one line of integer arithmetic stepped reversibly on a lattice and read only through clicks, with measurement as the meeting, at a NodeDetector, of the future with the past, the NodeDetector being the one place of the random draw; the forms it reaches, the continuum's known formulas in lattice form from that one line, gravity's vector sector among them under the one assumption, each with its inputs and its status (the list of Section 5.1), and the lattice's own, the terms that depart from the continuum or have no counterpart there, a prediction where no measurement has read them and bounded where one has (the list of Section 5.2); and the method of claim, every statement carrying its status and its fence, every input and every miss named, and the derivations, the files and the implementation public. Among the forms, one result is this paper's own: from the one assumption applied to a moving source, with no number put in, the coefficient 4 being the sum of the clock's and the ruler's exponents, follow the vector sector of gravity, frame dragging's form  $g_{0j} = -4U v_{s,j}$ ,  $\alpha_1 = \alpha_2 = 0$  and  $\eta_N = 0$  at nature's gap, where the write without the odd Link part gave  $\alpha_1 = -8$  and no drag (derived under the one assumption; clicks; Section 5.1; S.56).

The precursors, one line each: the physical world as a discrete computation [2], as reversible cellular automata [3], of which the present step is a second-order one in integers with one Euclidean division (the bit-reversible integer leapfrog with the remainder kept [4]), as Elze's integer-valued cellular automata with Schrödinger's equation as their continuum limit [5], where here the step is one Euclidean division whose remainder stays at the Node, nonlinear in the integers, two levels and a remainder with no amplitude, and the click layer with its one draw, Bialynicki-Birula's unitary cellular automata for the Weyl, Dirac and Maxwell equations on a lattice [6], the world as a cellular automaton [7][8], the unitary lattice automata [9], the event-based simulations with detectors that click [10], the closest prior work to the click, and as a deterministic foundation beneath quantum theory [11]. The time-symmetric accounts of measurement, Wheeler and Feynman's absorber [12], Cramer's transaction [13], the two-state vector formalism [14] and Wharton and Argaman's locally mediated reformulations [15], of which the click as a meeting is the lattice form, both directions one integer line exact to the bit and the choice among the meetings the one non-local element. The

exponential metric, Yilmaz’s scalar-field theory [16] and the position-dependent index on flat space [17][18] (the paper adds the route from Rule3 under the one assumption and the finite-gap factors), whose photon sphere at  $r = 2m$  and critical impact parameter  $2em$  [19, Section 7] are the shadow’s form here, the paper’s own being the route to that metric from Rule3. The Link phases as a lattice gauge field [20], and an integer random walk reproducing Schrödinger’s statistics [21], where here the lattice is deterministic and the one draw is the click’s [22][23].

Two answers stay: Feynman’s bound on a local classical computer [24] holds, the excess placed in the declared counting rule of the click and not in the lattice (Section 7); and a lattice’s Lorentz violation at order one unless its terms are tuned [25][26] is answered in part: the model has no loops, an absence listed among the gaps, and the band’s factors close with the gap, at a pair the file’s width does not reach (Section 8.1).

### 1.3 How to read the paper

Every claim of the paper carries its status in the parenthesis that closes its sentence, the status word first, then its fence, then its pointers, and the claims list at the front of the supplementary material (Online Resource 1) holds every claim with its status, its fence, what it rests on and its derivation; the status is one of eight words, all of which this paper uses, inspiration on the conjectures of Section 9 alone:

- *theorem*, proved for every lattice.
- *derived*, obtained from theorems by argument.
- *computed*, a number obtained from the rule’s lines by derivation or by a named run of the implementation.
- *assumption*, a line held by decision and named where it is used.
- *hypothesis*, a line outside the model, under its own name.
- *experiment*, a number the model does not fix, with the experiment named that would fix it, the paper performing none.
- *fitted*, an input chosen with a known result in view.
- *inspiration*, a direction named in the conjectures of Section 9 and in the long version at the tag, with no result claimed.

Beside them the sentences and the claims list use calibration (a number taken from nature), declaration (a universe file’s own number), definition, untested (no measurement has read it), no line (the law has no line that gives the quantity; nothing is derived and nothing contradicted, and the row names what a line would have to give), bounded (a measurement bounds the form and confirms nothing), a postulate (one of the model’s stated postulates, an assumption), a prediction (a lattice term no measurement has read), by construction and no test (an identity of the definitions that no measurement can refute) and a comparison (a measurement set beside the model’s number, a supporting finding and not a proof). Every formula carries its fence: *lattice*, a reading of the lattice’s own numbers that no NodeDetector reports, a diagnostic; *clicks*, a formula for what a NodeDetector reports, the only kind that can be set beside nature. A bare “computed” is a number from Rule3’s lines, the algebra evaluated; “computed, a run” names a fact of the one run the paper reports, made on the archive’s implementation named in the Statements and Declarations, which is not the paper’s subject. The algebraic steps of every derivation are in the supplementary

material (Online Resource 1), Derivations S.1 to S.56, cited as (S.*n*); the material the paper sheds beyond the supplement, the hypotheses under their own names and the openings for other worlds, stands in the long version of the paper at the archive's tag **paper-long-v1.1** [27], cited by its derivation number there, (S.*n* of the long version). The paper cites two documents of its archive, the law [28] and the decisions [29]. Every sentence that says what a thing is, light, mass, an atom, gravity, is a sentence about the model and its files, never about nature; the theorems are theorems about the rule, and every comparison with nature is a supporting finding and not a proof. The five coined words of the paper, Node (with its Links and Ports), NodeDetector, Rule3, click and pace, are defined at their first use; the long version's other coined words are given in standard terms, with the glossary at the supplement's front. A large language model was used as a tool throughout the work, as the Statements and Declarations state.

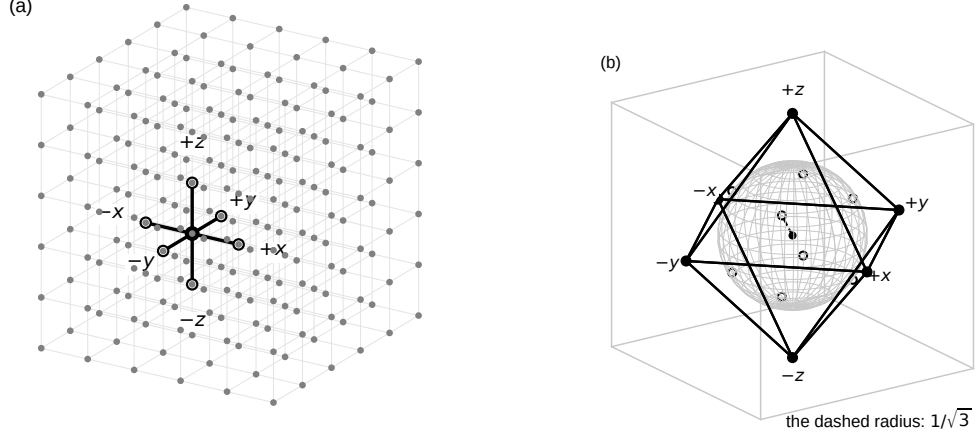
## 2 The model and the rule

The physical location is a *Node*, and its complete local information is its *NodeState*: for every field and every part (a part one level pair of a field: a potential field's time and axis parts, a bound state's own levels, a pair field's two lines), two integer levels, the present and the past, and one remainder, the potential fields' levels being the potential level a Node holds for the other fields to read; nothing else is kept at a Node, and every count, current, charge or stress is a reading computed from these numbers when it is needed. Nodes are joined by *Links* through six directional *Ports* per Node, and the cubic array of Nodes is the lattice. A *field* (the long version's family) is one kind of physical value with its rational pair and its shape, and its *configuration* is its levels over the lattice. A *pace* is the step factor of a Node (its clock) or of a Link (its ruler). A *NodeDetector* is the model's counter, a connected region of Nodes declared in the universe file, and its *click* is one quantum credited to it, the one measurement.

### 2.1 The lattice and its group of 48

The lattice is a box of  $N_x \times N_y \times N_z$  Nodes, each joined to its six neighbours by Links through its Ports  $\pm x, \pm y, \pm z$  (Fig. 1(a)). An axis is declared periodic, open (beyond an open face a field's rest is read, 0 for a field without one) or folded (one layer, whose two Ports return the Node itself); an inner face is a plane of Nodes declared closed except for declared gaps, a wall with slits; and a face may be declared receding (an assumption), the box growing by a layer of zeros whenever a nonzero level reaches the Node next to it, so that a run on such a box is a run on the unbounded lattice, with no cosmological redshift since nothing stretches.

The symmetry group of the cube has 48 elements, the 24 rotations and, with the reflections, the signed permutations of the three axes:  $O_h$ , the group of the octahedron of the Nodes one interval away (Fig. 1(b)), with the translations the space group of the simple cubic lattice. It acts by permuting the Nodes and their Ports and carries every level pair with its Node, and Rule3 commutes with it, since its line reads the six Ports symmetrically and names no axis. Under the group the form  $D$  of Eq. (1) is a scalar, even under the time reversal, which exchanges a field's next and before levels



**Fig. 1** (a) The lattice in space, a cubic array of Nodes, here  $6 \times 6 \times 6$ , each joined to its six neighbours by Links through the Ports  $\pm x, \pm y, \pm z$ ; one Node and its six neighbours, the Nodes one interval away from it, are marked. A change passes between neighbouring Nodes at one Link per interval, the causal bound, and light at  $1/\sqrt{3}$  of it. (b) The Nodes one interval away form the octahedron  $|x| + |y| + |z| \leq 1$ , whose six vertices are the six Ports; its symmetry group is the cube's,  $O_h$ , the 24 rotations and, with the reflections, the 48 signed permutations of the axes. The inscribed sphere, of radius  $1/\sqrt{3}$ , touches the eight faces on the diagonals of the cube, and that radius is light's speed in Link per interval. Drawn from the definitions; no run

and keeps its present, and carries the step to its inverse; the current  $F$  is a vector, odd under it; the Wronskian  $W$  is a scalar under the cube's 48, odd under the time reversal and under the plane's reflection; and the click is a scalar, naming a region and an interval, never a Node or an axis. No boost, no rotation beyond the cube's 24 and no scaling is in the group, so Lorentz's group and the continuum's rotations are no symmetries of the lattice, and what the paper finds of the moving clock it finds as a property of the clicks (theorem; lattice; S.4).

## 2.2 Rule3 and the vacuum band

The paper follows from one line of integer arithmetic, Rule3, and the inputs named beside it: one write, three readings, the one assumption of the forms (Section 2.3), the counting rule of the click (the long version's credit) and the declarations named where

they stand. The line holds at every Node, in every interval and for every field:

|                |                                                                                                                                                                                                                                                                                                                                     |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rule3:         | $w a_{\text{next}} + r' = \sum_{j \sim i} R_{ij} \text{arr}_j + S a_{\text{now}} - w a_{\text{before}} + r,$ $0 \leq r' < w = 6 \text{den} \Gamma^2, \quad R_{ij} = 2 \text{num} p_a(i, j)^2 = 2 \text{num} p_i^2 q_{ij}^2 / \Gamma^2,$ $S = 12 \text{den} \Gamma^2 - 12 (\text{den} - \text{num}) p_0^2 - \sum_{j \sim i} R_{ij};$ |
| the one write: | a potential field's part at a Node gains $\left( \sum_f w_f q_f(\text{Node}) + r \right) \text{div } E$ ,<br>the remainder $r$ kept there;                                                                                                                                                                                          |
| the readings:  | $D_i = \text{now}_i^2 - \text{next}_i \text{before}_i,$<br>$W_i = \text{re}_{\text{now}} \text{im}_{\text{before}} - \text{im}_{\text{now}} \text{re}_{\text{before}},$<br>$F_{ij} = \text{num} (\text{now}_i \text{before}_j - \text{before}_i \text{now}_j).$                                                                     |

(1)

In the first line  $a_{\text{now}}$ ,  $a_{\text{before}}$  and  $a_{\text{next}}$  are the field's level at the Node now, one interval before and one interval later;  $\text{arr}_j$  is the level of the neighbour  $j$ , the six Ports summed;  $r$  and  $r'$  are the remainder kept at the Node before and after the step;  $\Gamma$  is the Node clock;  $[\text{num}, \text{den}]$  is the field's pair,  $\cos \omega_0 = \text{num}/\text{den}$  its rotation per interval at wave number zero;  $p_0$  is the Node's clock pace,  $p_i$  its own pace and  $p_a(i, j)$  the pace of the Link through the Port, the Node's pace times the Link's factor  $q_{ij}/\Gamma$ ; at the vacuum's paces the two Links of an axis share one  $R_a = 2 \text{num} \Gamma^2$  and  $S = 0$ , and  $\text{div}$  is floor division. One whole number is formed and divided once by the wall  $w$ ; the quotient is the next level, the remainder stays at the Node, nothing is rounded away, and the step back is the same division (Theorem 1). The second line is the only way a potential field's level (the long version's holder: a field that holds a value for others to read) is ever written, the sourcing fields' quantities  $q_f$  at the Node, each a form  $D_f$ , a Wronskian  $W_f$  or a stress  $T_{aa,f}$ , read in the Node's own units (Section 2.3; S.34), weighted by  $w_f$ , summed with the part's one remainder and divided by the part's wall  $E$  (a divisor, not the form  $E$  of Section 3.3),  $E_s T$  for a time part and  $E_s W_c$  for an axis part. The third line is what is read from a configuration and never written: its form  $D_i$ , which sources the potential fields; its Wronskian  $W_i$ , its charge (for a plane field  $\text{re}$  and  $\text{im}$  are its two level pairs); and its current  $F_{ij}$  into the Node  $i$  from its neighbour  $j$ . A NodeDetector reads that current through its boundary in quanta of the field's unit  $W_c = 3 \text{den} T$ ,  $T$  the quantum's norm, one quantum's inflow being  $W_c \sin \omega_0$  in the current's unit and its energy  $T \sin \omega_0$  in the quantum's, and a massless field's quantum  $W_c$  itself.

At the vacuum's paces the line reads  $a_{\text{next}} + a_{\text{before}} = \frac{\text{num}}{3 \text{den}} \sum_a (\text{arr}_{+a} + \text{arr}_{-a})$ , whose rotation at the wave vector  $\mathbf{k}$  is  $\cos \omega = \frac{\text{num}}{3 \text{den}} \sum_a \cos k_a$ ,  $\cos \omega_0 = \text{num}/\text{den}$  at wave number zero. For the massless pair  $[1, 1]$  along one axis  $\cos \omega = (2 + \cos k)/3$ , so  $\omega = |k|/\sqrt{3}$  as  $k \rightarrow 0$ , and along the cube's diagonal  $\cos \omega = \cos k$  exactly, with no dispersion (S.1): light's speed at long wavelength is  $1/\sqrt{3}$  Link per interval (derived; lattice; S.1), below the causal bound of one Link per interval (assumption, postulate 4 of the long version's list, the step's reach; lattice), and its group speed falls at shorter wavelengths as  $1 - 0.083 k^2$ . In nature's names the vacuum's line is the discrete Klein-Gordon equation (a quantum walk's lattice form of it in [30]), exact but for the remainder term, with  $c_K^2 = \text{num}/(3 \text{den})$  and  $m^2 = 2(1 - \text{num}/\text{den})$ , the rest rotation the mass, and its slow limit at small wave number is Schrödinger's equation

with the inertia  $m^* = 3 \tan \omega_0$  under three named conditions (S.53);  $m^* c_m^2 = \omega_0$  by construction for every pair, with  $c_m^2 = \omega_0 / (3 \tan \omega_0)$  the band's kinetic scale, the model's  $E = mc^2$  read in three click numbers, nature's its small-gap limit, the departure  $\omega_0 / \tan \omega_0$ ,  $1 - \omega_0^2 / 3$  to the order  $\omega_0^2$  (derived; clicks; S.18, S.21).  $c_K$  is the discrete equation's coefficient at every wave number and  $c_m$  the band's kinetic scale at rest; they differ at a finite gap and agree at nature's (derived; lattice; S.1, S.18, S.21, S.34 (c)). At the vacuum's pace Rule3 is the leapfrog scheme of the Klein-Gordon equation, exact in the variable  $s = 2 \sin(\omega/2)$ ; what the law adds is one rational [num, den] that fixes mass and speed together, departures that are predictions with bounds (Section 5.1), the integers with the remainder, and the paces.

### 2.3 The paces and the one assumption

A potential field enters a field's step by changing the field's paces at a Node, the coefficients of Rule3 there,  $c_i$  the potential level the field reads at the Node (the long version's content),  $U = c/\Gamma$  the same in the potential's unit, and  $t_a(i, j)$  the Link's stress (the long version's tension, the level the Link holds), one number per Link from the parts at its two ends (S.7; the Link's number the mean of its two ends' readings, S.33):

$$\boxed{\begin{aligned} p_0 &= \Gamma (1 - 1/\Gamma)^c, & p_i &= \frac{p_0 (c_i)^2}{\Gamma}, & p_a(i, j) &= \frac{p_i q_{ij}}{\Gamma}, & q_{ij} &= \Gamma - t_a(i, j), \\ c &= \sum_{\text{reads}} \text{weight} \times \text{the read field's time level.} \end{aligned}} \quad (2)$$

A positive weight makes a hollow, an attraction, and a negative weight a hill; every field reads the potential fields its declaration names at the weights it names, a plane field also the charge potential, and sources what it reads, so every pair of fields that read each other is reciprocal.

Nothing in Rule3 fixes how a level enters the paces. Beside the line, the model holds that a potential level is a count of units and that every act is measured in the Node's own units,  $hN = 1$  in its units form: the one assumption of the forms (assumption, fitted; the composition derived from it; lattice; S.34). From it the clock composes and a uniform potential level  $U$  changes only the units, time by the lapse  $N = p_0/\Gamma$  and length by  $h = p_0/p_a$  per Link (S.34). One sentence is not in Rule3: a Link is clocked twice by its reading Node's own clock,  $p_i = p_0^2/\Gamma$ , the ruler going with the clock. The lattice says why twice, since a Link has two ends; it does not say why the reading Node's clock stands at both ends rather than the product of the two ends, which gives  $\beta_K = -1$  (the post-Newtonian  $\beta$  as the orbits read it, Section 5.1). The sentence is one discrete choice: the one assumption,  $hN = 1$ , is fixed by the bending's measurement,  $4U_b$  for "twice" ( $\gamma = 1$ ), and the perihelion's 43 arcseconds choose the Node's own reading of the Link's pace over the two ends' product ( $\beta_K = -1$ , 72 arcseconds) and then stand as the test of the choice,  $\beta = 1$  as the gap closes,  $\beta_K = 1.25$  at the computed pair, where the band's  $k^4$  term adds 1/8 of the advance to first order for an orbit in a coordinate plane, 13 to 15 percent by the ray integration (S.19); it is the first of

three choices made with a known result in view (Section 8.2); every gravitational form of Section 5.1 is conditional on it, the bending's measurement its calibration and no test of it, the perihelion the test of the end's choice, Shapiro's delay and the redshift's factor  $f(\omega_0)$  its consequences, read against Cassini and Pound and Rebka, and the shadow's  $2em$  bounded by measurement and not confirmed.

For a moving source the potential's flux writes three odd Link levels  $V_{ij}$ , the lattice's  $g_{0i}$ , read as a phase by plane (charged) configurations and by no real one. The same assumption holds for a source that moves: the field's count flux sources gravity's three odd lines as its count sources the time line, in the count's unit,  $V_{ij} = -V_{ji}$ , and the Port turns an arriving plane configuration by the phase  $\theta_{ij} = -\theta_{ji} = 4V_{ij}\omega_0$  of its field, the 4 the clock's and the ruler's exponents at  $\gamma = 1$  and no number put in, paid inside Rule3's one division, the odd Link part of the metric, the clock family's three odd parts; a one-part configuration, light among them, carries no phase (assumption, the one assumption for a moving source; the coefficient derived under it; lattice; S.56). Under it the model's static metric for light is  $ds^2 = -e^{-2U}dt^2 + e^{2U}d\mathbf{x}^2$ , to  $O(U/\Gamma)$  and in the ray limit, the exponential metric [16], Einstein's to the first post-Newtonian order for a static source,  $\gamma = \beta = 1$ , the frame's parameters derived under the same assumption for a moving source (Section 5.1); the clock's rate is  $N$ , exactly in the lattice's variable  $2\sin(\omega/2)$  and in  $\omega$  up to the finite-rotation correction of order  $\omega_0^2/12$ , 6.3 percent at the computed pair as the exact  $f(\omega_0) - 1$  and below  $1.1 \times 10^{-20}$  at nature's gap under the photon bounds of Section 5.1 (derived under the one assumption; lattice; S.34 (c), S.31 for the clock's rate).

From the coefficients, the rotation of a configuration at a Node with clock pace  $p_0$  and Link paces  $p_a$  is (S.1)

$$\cos \omega = 1 - \left(1 - \frac{\text{num}}{\text{den}}\right) \left(\frac{p_0}{\Gamma}\right)^2 - \frac{\text{num}}{3 \text{den}} \sum_a \left(\frac{p_a}{\Gamma}\right)^2 (1 - \cos k_a). \quad (3)$$

The mass term scales with the clock's square and the band term with the square of the Link's pace: at  $k = 0$  the rest rotation of every massive band obeys  $1 - \cos \omega_b = (p_0/\Gamma)^2(1 - \cos \omega_0)$ , and along a Link the speed of every wave falls with the Link's pace, so light is delayed and bent by twice the clock's share, its speed at a Node of potential level  $x$  being  $N^2 = e^{-2x}$  times its vacuum speed, an index  $n = e^{2x}$  (S.2). A pace must lie at or below the guard's edge  $P$ ,  $P^2 = 2 \text{den} \Gamma^2 \text{div}(\text{den} + |\text{num}|)$ , the Courant, Friedrichs and Lewy condition of the explicit scheme that Rule3 is at the vacuum's pace, checked once at load, light's vacuum pace sitting at the edge at the level 0, where S.3 admits the linearly growing checkerboard mode, and below it at the files' level 60 (theorem for the edge; lattice; S.3); under the composed paces no Link closes in the rational paces (derived for the closing; assumption for the frozen Node's energy; lattice; S.39), since the clock is positive at every level, so a deep well freezes a clock and closes nothing, and a Node whose Link pace rounds to 0, at  $(\Gamma/2) \ln 2\Gamma$  levels, is a frozen Node, read by its neighbours, reading nothing and dark (S.3, S.39).

## 2.4 The interval and the clock

A reading writes nothing on the lattice, and a static configuration (now = next = before) has no form and no current, so it carries no quanta and no NodeDetector sees it. Every read takes the values at the interval's start, so the interval's dependency radius is one Link; its four acts, in order, are the read of the paces, Rule3 on every configuration, the readings with each NodeDetector's report, and the write of each potential field; after the four acts, at a realised click, the NodeDetector's one write (Section 2.5). The conservations are per field, and the lattice conserves no total, since the quantum share (the field's count in a region, Section 3.4) moves with the paces and no current compensates; the click layer has a count register of its own, one integer per configuration that clicks alone move (the long version's books), and neither layer keeps a total energy (theorem for the step's conservations; assumption for the clicks'; lattice, clicks; S.49). The line and its clock are fixed: every mend of the model is a field, a potential field, a declaration or a read, never a change of Rule3 (an assumption).

## 2.5 The click: the meeting and the NodeDetector

*Assumption 1* (the click is a meeting, the law and not a hypothesis) A field's configuration runs on the lattice in both directions by the one line: forward from the source Node of its initial data, the possibilities, the future; backward from the realised click, what happened, the past, Rule3 at  $-1$ . A click is the meeting of two quanta at a declared NodeDetector: the quantum of the incoming configuration run forward, the future, and the transition of the configuration that waits for it, the click's unit, the past read back from it. Its quantum share is their product at the region, at resonance (Section 4.3): the form  $D = \text{now}^2 - \text{next} \times \text{before}$ , the now against the future times the past. Which meeting is realised is the NodeDetector's random draw, the world's one (a bound state's NodeDetector's own seed), and the lattice draws nothing. The one non-local place is the counting rule's one draw over the NodeDetectors that read one configuration, through the source Node. The NodeDetector only asks whether the future and the past met at it.

There is one kind of reader, and the universe file decides what it holds: with no configuration of its own a NodeDetector reads the field that arrives at it, a bolometer's boundary, as the two slits' screen does; given a configuration of its own, with parts, transitions and a lifetime, it stands for an atom or an ion and emits as well. Algebraically, the step  $S_t$  is a bijection of the state space (Theorem 1), its forward orbit the future and its inverse orbit the past; a NodeDetector  $R$  is a declared region whose boundary Ports and coefficients define  $Q_R$ , the inflow over the counting window, and a click is the pair  $(R, t)$ , one whole quantum credited to  $R$  at the interval  $t$ , drawn with the probability  $s_R / \sum_{R'} s_{R'}$ ,  $s_R = Q_R(x_t) / W_{\text{rec}}$ ,  $W_{\text{rec}}$  the configuration's energy per quantum (Section 3.4),  $W_c$  for light and  $W_c \sin \Omega$  for a born quantum (S.46); the decision is the one map in the model that is not a bijection. The parts of a configuration are read with the NodeDetector's coefficients, each part paired with the same part through the common source Node, then summed and squared; the two slits are the case with nothing to pair, Bell's check the case through the common source Node (Section 7), the joint energy computed from the forward configurations alone by the linearity of the rule. A NodeDetector with Nodes alone counts by the half-up rounding

of the total over every region reading the configuration, so one region's count moves with the others' inflows by at most one quantum per window, the one named non-local place; a bound state weighs each transition by its own transition's quantum share, so its marginal is its own whatever the other bound states' (S.57 (b) of the long version [27]).

*Assumption 2* (the NodeDetector's declaration, its decision and its write) A NodeDetector is one connected region declared in the universe file, two Nodes or more and at least half the wavelength of the field it reports across the beam (4 Links at  $\lambda = 8$  Links), one Node or more where the file gives it a configuration of its own. Each interval it reads the inflow through its Nodes' boundary Ports, Rule3's currents summed in integers into its counting window, declared in the file and never read off the wave, which opens at its last write and closes at its draw; its click is its report, one line per quantum (the window's index, the NodeDetector's name, the field and the part, the count before and after, its own clock), never a Node, one unit of the field,  $W_c \sin \omega_0$  of the inflow's energy, being one unit of count. The counting rule credits one quantum to one NodeDetector, drawn with the world's one random draw by the energies of the inflows over every declared NodeDetector of the file, one credit per quantum and never two, the run deterministic per seed. After a realised click the NodeDetector writes once, at the one Node of its region that the same generator's next draw picks by the inflow counted through each Node: one whole quantum of the arriving field is absorbed there, its count in the register falls by one, the absorbing configuration's part rises by one, laid in its mode with the count's own scaling,  $A^2 = (N+1) T / (2 \sin \Omega)$ ; where the arriving field's energy at the drawn Node is at most  $W_{\text{rec}}$  its levels and remainder there go to 0, and a denser field is undepleted, its count moving in the register alone and the deficit printed. Nothing is scaled by a factor and no number enters, and the change spreads by Rule3 as an event (Section 2.5.1). Backward, to the past, a reading and never a write: the counting rule's reading through the source Node, which chooses which tree is the past, the one non-local place. A NodeDetector given a configuration of its own, held by its declaration and never bound by a well, emits as well as takes, one whole quantum of light laid at the drawn Node at  $\Omega = \omega_e - \omega_g$ , drawn per window at the rate, the beat's current over  $W_c$  plus the rate's declared constant (S.46).

What produces a click is deterministic, the meeting; which possibility becomes the click no line of the law gives: the NodeDetector draws a place and a combination, never a value and never a path. The generator is a declared linear congruence of the full period [31], so Born's rule on the lattice [32] is the proportionality of the weights to the energies, a declaration of the NodeDetector and not a finding of the model, each outcome's frequency within  $2^{-\text{width}}$  of its weight's share and not exactness (S.46); the successive draws' independence is assumed and not proven, and the scatter  $\sqrt{NP(1-P)}$  is conditioned on it. Exclusivity, one click per quantum, is the counting rule's declaration, which gives antibunching [33], one quantum and one click per configuration between far NodeDetectors; under it no signal passes (derived under the credit's one draw, the NodeDetector's declaration, for exclusivity; derived for antibunching and the one form; assumption for the draw, the write and the counting rule; clicks; lattice for the write; S.46); and the NodeDetector's resolutions are its own declarations and no uncertainty relation of the model's own, in place the region ( $\Delta y k \geq \pi$  at half a wavelength), in frequency the counting window ( $\Delta \omega \geq 2\pi/n$  over  $n$  intervals) and in count the draw ( $\sqrt{NP(1-P)}$  over  $N$  clicks); the dead time is

not derivable, a declaration: the quantum shares credit a region any whole number of quanta in one interval, and  $W_c$  over the inflow's rate is the mean spacing of the clicks and not a least one (declaration; derived under it; clicks; S.8, S.46). The lattice holds only the future and has no arrow, Rule3 being a bijection at every interval; the clicks have the arrow, each forward only, the one act outside the reversal: the direction of time belongs to the clicks and not to the lattice (theorem for the step, the arrow the NodeDetector's declaration; lattice; S.46). A quantum changes field or part only at a click's write, and between clicks every count is constant (derived, two derivations; clicks; S.46). Light is the charge potential's own configuration, sourced by the Wronskian of the fields that read it, stepped by Rule3 at the paces of its reads and written by the one write: a static rotating bound state writes a static level, Coulomb's field, and radiates nothing; a breathing or accelerating one writes a varying source that the charge potential carries away, the spreading events of Section 2.5.1, the light (derived; lattice for the Wronskian and the source, clicks for the far NodeDetector's rate; S.40); no configuration reads its own write of the charge potential.

*The steps up to a click, with two NodeDetectors.* At A a window closes: the draw picks a place and a combination, the write moves one count, A's part down by one and one quantum of light born at the beat, its wave vector set. From A's Node the quantum spreads by Rule3 as events, a fan of possibilities inside the causal cone at one Link per interval, light's front within it at  $1/\sqrt{3}$ , the wave vector the phase step between neighbours; nothing travels as a point. At B the inflow is read at its Ports and the shares accumulate; the configuration run forward from A meets the transition that B's click reads backward, Rule3 at  $-1$ : the meeting. At the window's close B draws whether and which future continues, the NodeDetector and the combination, under Born's weights of the shares; its write's Node is the generator's next draw. B's write moves one count, the light's down by one and B's part up by one, at the drawn Node; the drawn future is now a past. From B's Node the erasure runs over the rest of the fan (Section 2.5.1). Between the two clicks one count has moved twice and one wave vector, set at A, has been carried by Rule3 and read at B, locally at every Link; the one thing not local is B's draw, one per configuration over every NodeDetector that reads it (assumption for the draw, the write, the counting rule, the erasure and the empty wave; derived for the rest; lattice, clicks; S.46).

### 2.5.1 How an event spreads

On the lattice there are events and amplitudes. An event is a level split by Rule3 to the six neighbours. An amplitude is the level Rule3 produces at a Node, the sum of the split events there. The configuration, its levels over the lattice, is the amplitudes. In Universe24 there are clicks, the NodeDetectors' reports, and nothing else. An event spreads by Rule3: the level splits to the six neighbours along the Links, one Link per interval, the causal bound of the line, which reads the six Ports and nothing farther; light is such spreading events at  $1/\sqrt{3}$  Link per interval (S.1), and a click's erasure runs at the full bound inside its cone. The causal front after  $d$  intervals is the octahedron of the Link distance, and light's front at long wavelength is the sphere of radius  $d/\sqrt{3}$  inscribed in it; the two touch at the faces' centres, on the four diagonals, where the band is  $\cos \omega = \cos k$  exactly, and along the axes the bound runs  $\sqrt{3}$  ahead of the

light (assumption, postulate 4, for the bound; derived for the two fronts; lattice; S.1). In the two slits the initial data's events pass both gaps, every path of the future at once; the fringes are where split events from the two gaps meet at the screen with the same sign. The past is made of the same split events: read backward from the realised click by the same line, Rule3 at  $-1$ , they form the past's tree, which also passes both gaps back to the source Node. What stays on the lattice, the amplitudes no click realised, is not in Universe24. After a click they are a cancelled future at three levels, the counting rule's, the NodeDetectors' and the lattice's: in the count register the configuration's count is at 0 and no NodeDetector credits it again; where they reach a NodeDetector backed by a declared face, their quantum share leaves the lattice there; on the lattice the absorbed configuration is erased from the click's Node outward at the causal bound, one Link per interval, inside the click's causal cone ( $\sqrt{3}$  times light's speed), the ball of radius  $d$  holding  $(2d+1)(2d^2+2d+3)/3$  Nodes of zeros, never faster, and beyond the front its levels remain, an empty wave with count 0, stepped on and never credited, that still sources the potential fields it writes to until the front arrives (an assumption, the NodeDetector's declaration; lattice, clicks; S.46).

### 2.5.2 The click's algebra beyond the law

A click is the one act of the law that moves a register's count,  $-1$  for the arriving configuration and  $+1$  for the taker's part, the taker the configuration whose transition the click reads or the quantum leaving the lattice at a declared face, every other act leaving every count unchanged (assumption, the NodeDetector's declaration, for the exchange; every other act leaving every count unchanged derived, Theorems 3 and 4; lattice, clicks; S.45); the draw chooses whether and which future continues, a place and a combination, never a value, a path or the time (assumption, the counting rule; clicks); the write at the drawn Node changes the share by the booking identity of the list of Section 5.2 (derived; clicks; S.46), keeps the charge, balances the frequency by the beat (Section 4.3) and conserves no momentum, the quanta laid with a phase and no wave number (Section 8.1); a quantum's momentum is a chain, its frequency set at its birth click by the beat and its wave number by light's band, its direction none at the one-Node lay, carried by Rule3, conserved in vacuum and turned in wells by the index with the frequency kept, and read at its death click (derived; lattice; S.2, S.7, S.20). Under one proposition, a line beyond the law's write of Section 2.4, under its own name, chosen with a known result in view, that nature's clicks conserve momentum event by event [34], and used by no run of this paper (hypothesis, fitted), the write carries the arriving wave vector as a phase twist of the taker's configuration,  $\delta k_a = \text{sgn}(k_a) (|k_a| \text{div } M)$  per Link along each axis for a taker of count  $M$ , the remainder carried, the wave vector read at the region's Ports from the Link current and no number beyond its own reads (S.55 (c)). Derived under it in S.55: the charge kept and the integer wave vector conserved across the click,  $r_a + k_a = M \delta k_a + r'_a$  at every click, the books before and after, the twist written on the taker and the remainder carried (derived under the proposition; lattice; S.55); the recoil  $k^2/(2Mm^*)$  in rotation with  $m^* = 3 \tan \omega_0$ , vanishing for a large taker (derived under the proposition; lattice; S.55); Compton's shift in the non-relativistic regime [35] with the factor  $\omega_0/\tan \omega_0$ , 0.752 at [2, 3] from the exact factor, 0.778 to first order, the beat rule's number (S.55

(f)), and 1 at nature's gap, against 0.5015 at [2, 3] under the click's energy  $T \sin \omega$ , the lattice's own term at a finite gap, where Compton's formula is the Link's definition by construction, no test (derived under the proposition, nature's by construction and no test; lattice at the computed pair; S.55 (f)); and in a well of index  $n = e^{2U}$  the wave vector passed per click, the twist,  $n$  times the vacuum's at the same frequency, Minkowski's form, and the recoil in rotation  $n^2$  times it (derived under the proposition; lattice; S.2, S.55). The paragraph is algebra on the law's definitions; no run of the paper rests on it; S.55 carries the algebra, its checks and the integer threshold of the twist.

### 3 Exact properties

This section states what holds exactly on every lattice, each item proved in S.45, Theorem 4 in S.5 and S.6: the reversibility of the rule, the kind of function its acts can form, the conserved form with its current, the count as a reading of the configuration (S.5, S.6), the Wronskian as the charge (S.9) and the three exact bands.

#### 3.1 Reversibility

Let  $\sigma$  be the direction,  $+1$  forward and  $-1$  backward,  $\text{div}$  and  $\text{mod}$  the floor division and its remainder in  $[0, w)$ , and  $\rho$  the remainder carried in. With  $u = \sigma(\sum_j R_{ij} \text{arr}_j + Sx - wy) + \rho$ ,  $z = \sigma(u \text{ div } w)$  and  $\rho' = u \text{ mod } w$ , the forward call with  $(x, y, \rho) = (a_{\text{now}}, a_{\text{before}}, r)$  gives  $(z, \rho') = (a_{\text{next}}, r')$ , and the backward call with  $(x, y, \rho) = (a_{\text{now}}, a_{\text{next}}, r')$  gives  $(z, \rho') = (a_{\text{before}}, r)$ .

**Theorem 1** (the direction) *The backward call returns  $(a_{\text{before}}, r)$  bit for bit; the reversibility of every step needs no second function, the odd lines' source recomputed by the backward step as the paces are (S.56 (b)). (theorem for the step and where the writes change no level; the write an assumption; lattice; S.45)*

A NodeDetector's write is outside the reversal, as the initial data are; the whole run goes back,  $N$  intervals forward and  $N$  back returning every array of every field bit for bit, provided every number an act read at an interval's start is one the state after the interval still holds. On the two slits' file the run forward and back returns every array bit for bit over 40 intervals, the counting window opening at 45, so the check contains no click (computed, a run, the archive's `blind_and_reading.md` at 1fe3790a; lattice). The condition rules out one read: a coefficient read from a Node's own state across a step cannot be recovered at the step back, so a configuration's rotation sense is never read at a Node, and the charge potential is read by the configuration's shape, a plane or a line (Section 3.5).

#### 3.2 What the acts can form

Rule3's line is one act, and declared coefficients make four special cases of it, which the law also calls acts: the read, the line with its six arrivals; the one-Node step, the arrivals declared 0, which rotates a level by Chebyshev's recurrence; the division,

$R = S = 0$ , Euclid's division with the remainder kept; and the load, a declared integer in the numerator, which is how a count enters a level.

**Theorem 2** (what the acts can form, on the infinite lattice, for one field at fixed coefficients)  
*Every fixed finite composition of the line's four acts with fixed coefficients is an affine form of one field's levels plus a term bounded by a constant of the composition, since a floor is its argument less a fraction in  $[0, 1)$ ; so no product of two of its levels is written by the step on the whole lattice. The products across fields enter through the coefficients the read turns from the held levels: the read multiplies. On a bounded domain every integer function is a finite sum of floors of affine forms, and the theorem says nothing there. (theorem; lattice; S.45, S.56 (c))*

So every product of two levels is a reading and never a write by the step, and the readings of Eq. (1), the stress and the share are the only products the law takes. A line enters the law only if it passes three tests, generic (one primitive with declared integers, no field's name), vector (an act on the whole state, with no root and no floating-point number) and local (its own configuration and the six neighbours); a line that needs more is a hypothesis under its own name.

### 3.3 The conserved form

The read coefficient  $R_a(i \rightarrow j) = 2 \text{num } p_i^2 q_{ij}^2 / \Gamma^2$  carries the Node's factor and the Link's; where the paces differ the Node's factor is not symmetric and the Link's is, so the conserved form carries the weight  $1/p_i^2$  on its Node terms and the Link's factor on its Link terms:

$$E = \sum_i \frac{w(\text{now}_i^2 + \text{before}_i^2) - S_i \text{now}_i \text{before}_i}{p_i^2} - 2 \text{num} \sum_i \sum_{j \sim i} \frac{q_{ij}^2}{\Gamma^2} \text{now}_i \text{before}_j. \quad (4)$$

**Theorem 3** (the conserved form) *At constant paces for which a positive Node weight  $d_i$  with  $d_i R_a(i \rightarrow j) = d_j R_a(j \rightarrow i)$  on every Link exists,  $E$  with the weights  $d_i$  is exact under the line without the rounding. Under the law's read coefficient, with  $q_{ij} = q_{ji}$  and  $p_i = p_0(c_i)^2 / \Gamma$ , the weight  $d_i = 1/p_i^2$  cancels the Node's factor on both sides of every Link. So the weighted form, with  $\mathbf{M}$  the matrix of Rule3's step,  $\mathbf{a}_{\text{next}} + \mathbf{a}_{\text{before}} = \mathbf{M} \mathbf{a}_{\text{now}}$ ,  $\mathbf{D} = \text{diag}(1/p_i^2)$  and  $\mathbf{D}\mathbf{M}$  symmetric at every interval, is exact for any static potential levels and static Link stresses, and for a plane configuration under the Port's phase, the weighted coupling Hermitian, the same form exactly (S.56 (a)). Its Link term carries the Link's factor, so the current of  $E$  through the Link  $ij$  is  $2 \text{num} (q_{ij}^2 / \Gamma^2) (\text{now}_i \text{before}_j - \text{before}_i \text{now}_j)$ , twice the quantum share's current of Theorem 4, the plain current  $F_{ij}$  per Link's factor being the share's where no stress stands; the form with the plain current holds only where no stress stands, and with a stress it is not conserved (computed, two witnesses; lattice; S.32). (theorem at static paces, weighted; lattice; S.5, S.7, S.32, S.45)*

With the rounding, each Node's remainder adds the term  $(r - r')(\text{next} - \text{before}) / (w p_i^2)$  to the walk of the step's own two-level form  $Q$ , the form of now and

before without the wall (derived, exact in integers, two derivations; S.5).  $E$  is a diagnostic; where the paces move it changes exactly in the rationals by  $E(t+1) - E(t) = \mathbf{a}_{\text{next}}^\top \mathbf{D}(\mathbf{M}(t) - \mathbf{M}(t+1)) \mathbf{a}_{\text{now}}$ , 0 at paces fixed in time and negative where a well deepens under a standing configuration (a theorem by substitution). The momentum density along an axis is the current per axis, and its change is a stress reading on the Links, the momentum flux at uniform coefficients the wave's own stress (theorem for the line, derived for the momentum flux; lattice; S.7), a plane wave of amplitude  $A$  at the wave number  $k$  having the stress  $T_{aa} = \text{num } A^2 \sin^2 k \geq 0$ , as a wave's pressure is, so that light gravitates by its pressure too, written into the Link's factor and not the clock.

### 3.4 The count is the configuration's quantum share

Three things are called a count: the register's count  $N$ , an integer moved by clicks alone; the quantum share (the long version's share: the field's count in a region, Theorem 3's form over one quantum's norm),  $\sum_i e_i / W_c$ , a lattice reading that drifts between clicks by the currents and the remainders within the floors; and the quantum  $W_{\text{rec}}$ , the configuration's energy per quantum read once at its initial data, constant along its passage by the adiabatic invariant (S.14). At every Node the configuration's quantum share is

$$e_i = \left[ w(\text{now}_i^2 + \text{before}_i^2) - S_i \text{now}_i \text{before}_i - \text{now}_i \sum_{j \sim i} R_{ij} \text{before}_j \right] \text{div } (2p_i^2), \quad (5)$$

the floor of the rational term  $\tilde{e}_i$  of the conserved form at the Node with its weights; the counting rule reads  $\sum e_i$  in quanta over  $W_c$ , rounded half up to the nearest whole. At the vacuum's paces it is exactly  $3 \text{den}(\text{now}_i^2 + \text{before}_i^2) - \text{num now}_i S_6(\text{before})_i$ , with  $S_6$  the six-neighbour sum.

**Theorem 4** (the count is the configuration's quantum share) *Over one step of Rule3 the rational quantum share at a Node changes by exactly the six currents into the Node, each weighted by its Link's factor, plus Rule3's own rounding:  $\tilde{e}_i(\text{next}, \text{now}) - \tilde{e}_i(\text{now}, \text{before}) = \sum_{j \sim i} (q_{ij}^2 / \Gamma^2) F_{ij} - (\text{next}_i - \text{before}_i)(r'_i - r_i) / (2p_i^2)$  (S.5). So over any region the total energy changes only by the weighted currents through the region's boundary, the remainder terms and the NodeDetectors' floors, the floors each a residual in  $[0, 1)$  per Node and level pair. The total quantum share of any lattice whose Links read the same from both ends and whose every beyond reads 0 (the archived files, whose massless potential rests at 60 beyond the open faces, lie outside this condition) is non-negative for  $|\text{num}| \leq \text{den}$  at the vacuum's coefficients, with equality for a static uniform level of light and, on an even periodic lattice, for its checkerboard (S.6); one Node's energy can be negative at the initial data's edge, and a region's by at most half a level-square per boundary Port. (theorem under the two conditions; lattice; S.5, S.6)*

Over the whole passage of one forward wave packet the inflow accumulated at a region's boundary equals the region's final energy plus what left through its back, each bounded below by the theorem, so the inflow is non-negative up to the rounding and the counting rule by the energies is well defined; where a counting window is cut

inside a passage the NodeDetector credits  $\max(s_R, 0)$ , the floor at 0 the counting rule's declared rule (an assumption).

### 3.5 The Wronskian as the charge

A field of quanta laid as a plane carries its configuration as two level pairs, (re, im), stepped by one call of Rule3 with one set of coefficients, each with its own remainder. The charge density at a Node is the reading

$$W_i = \text{re}_{\text{now}} \text{im}_{\text{before}} - \text{im}_{\text{now}} \text{re}_{\text{before}}, \quad (6)$$

the Wronskian of the two pairs. The line without its division conserves it exactly, with the weights of the conserved form, under the condition of Theorem 3 on the paces (theorem for the conservation; lattice; S.9); the integer step adds to each part the remainder term  $\epsilon = (r - r')/w$ ,  $|\epsilon| < 1$ , so the integer  $W$  changes at a Node by  $\epsilon_{\text{re}} \text{im}_{\text{now}} - \epsilon_{\text{im}} \text{re}_{\text{now}}$  per interval, and an exactly conserved integer charge is open; the weighted total  $\sum_i W_i/p_i^2$  is exact at any paces fixed in time. A real configuration has  $W = 0$ , so light, the real configuration of the charge potential, is neutral; a rotating configuration has  $W$  of either sign by its sense, and antimatter, in the model's words, is the configuration of the opposite sense, the same field, gap and inertia with the opposite charge,  $q = \mp Ce$ , which sense is nature's matter being the one declared bit (derived for the opposite sense, the bit declared; clicks for antimatter; S.33, S.41, S.52). The charge potential does not read its own level, so light does not read it and a static charge does not bend light.

### 3.6 The three exact bands

The line of one Node at rest,  $a_{\text{next}} = 2 \cos \omega_0 a_{\text{now}} - a_{\text{before}}$ , is exact in integers only where  $2 \cos \omega_0$  is an integer, so the rule carries exactly three exact massive rotations,  $2 \cos \omega_0 = 1, 0$  and  $-1$ , with the periods 6, 4 and 3 and the pairs  $[1, 2]$ ,  $[0, 1]$  and  $[-1, 2]$  (theorem; lattice: an integer  $2 \cos \omega_0$  strictly inside  $[-2, 2]$  is 1, 0 or  $-1$ ). The rule allows the bands  $[0, 1]$  and  $[-1, 2]$ , and this world does not use them.

## 4 The fields, the bound states and the clusters: what the algebra builds

The algebra builds three things from the one line: fields, each a band of the rule named by its pair; bound states, each a configuration standing in a potential well; and clusters, bound states standing in one another's wells.

### 4.1 Fields, potential fields and the universe file

A field is a band of the one rule, not a separate law, and its entry in the universe file holds only its pair [num, den] and its shape, the dimension of a field of quanta, 1 for one real line or 2 for a plane, or the sources of a potential field, the form, the form with the three stresses, or the Wronskian, with its divisor  $E_s$  and, for the massless

potential, its rest; a field of dimension 3 is declared by no file and enters only as a hypothesis under its own name (S.37 of the long version, **paper-long-v1.1**). Under that hypothesis the proton is one quantum of a field of dimension 3, three real lines with the rotation among them, the quarks its three lines, no quark field declared, no count per line and no click of one line, and the neutron three real lines without it, the conversion the change between them; the configuration's indivisibility under the click, confinement in the model's words, derived under the hypothesis from the declared row; the baryon number, the flavours, the quark masses and the fractional charges are not derived and not claimed (a hypothesis under its own name; confinement derived under it from the declared row; no number of the paper resting on it; clicks; S.37 of the long version, **paper-long-v1.1**). When a pair is written over the lattice's one denominator  $\Gamma$ , a mass is one integer  $m$ , with  $\cos \omega_0 = m/\Gamma$  and the wall  $6\Gamma^3$ ; a particle of nature is read as one quantum of a field, the electron of the matter field and the proton of its own, and a field's gap, its mass, is declared and not derived. What a field carries is set by what Rule3 reads, the clock's pace and the three Link paces, so a potential field carries one time part, three axis parts and three odd Link parts,  $1 + 3 + 3$ , the last the potential field's odd lines, sourced by the field's count flux and read by plane configurations alone (Section 2.3; the shapes and the reads of this world's fields in Table 1). For a one-part real configuration no read of a current exists within the three tests, so light and a one-part line are not dragged (S.56 (h)).

Gravity enters in four steps, with no metric, no force and no law of motion: every field of quanta writes its form over the action,  $D_i \text{ div } T$ , into gravity's potential over the divisor  $E_g$ ; gravity's potential,  $[1, 1]$ , steps by Rule3 at its pair, resting around a static source at Poisson's solution,  $3G_0(r)s$ ,  $s$  the bound state's write per interval over the divisor and  $G_0$  the lattice's Green's function (S.37), Poisson's rest far away (Section 5.1); every field reads the potential's level into its own coefficients, Eq. (2), the clock's pace once and each Link's pace twice; and gravity's potential field carries three odd lines beside its time line, sourced at the matter's Links by the count's flux,  $V_{ij} = -V_{ji}$ , and resting around a moving source at  $\mathbf{V} = U \mathbf{v}_s$  as the time line rests at Poisson's, which a plane configuration reads at the Port as the phase  $\theta_{ij} = 4V_{ij}\omega_0$ , the delay of a rotating record, and a one-part real line does not read (derived under the one assumption; lattice; Section 2.3; S.4, S.37, S.48, S.56). What general relativity calls curved space stands here as the Link's pace, and what it calls the dragging of frames as the Port's phase; no space is bent in the model. A potential field's rest falls outside a bound state by  $e^{-\kappa}$  per Link,  $\kappa$  the inverse reach, the field's own band at zero rotation, so the gap is both the mass of the potential's quanta and its inverse reach, Yukawa's form [36], with the range of the list of Section 5.1 (derived; lattice; S.11, S.37). The write of a configuration is laid into two potentials (assumption, fitted, for the pair and the divisor; declaration for the range; lattice): the binding potential, the gapped pair [2400, 2401] at the divisor 1, its range  $R = 20$  Links following from the pair (S.11), which binds the bound states and is not the strong force; and the massless potential  $[1, 1]$  at the divisor  $E_g$ , gravity, long-ranged and weak by  $E_g$ , declared large in the archived files so that gravity enters no bound state's binding, nature's  $E_g$  read from  $G$  as  $k$  is read from  $\alpha$ . The massless potential rests at a declared level everywhere, the vacuum level 60 at  $\Gamma = 6,000$  (the long version's vacuum content, the level in the

**Table 1** The rule’s own universe, its four fields as the file `examples/events/rule.json` holds them (derived from the two keys of an entry, its pair and its shape; lattice): each field with its pair (in brackets over  $\Gamma = 6,000$ ), what it holds at its divisor, its shape, what it reads, and its free number. A further field, the pair field of light, two real lines never summed, is declared by Bell’s file (Section 7)

| Field                                                                                                                | Pair                                  | Holds, at the divisor                                                                                                                                            | Shape                                                                                                                              | Reads                                                                                          | Its free number                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| gravity, the time part and the three stresses                                                                        | $[1, 1]$<br>$[(6000, 6000)]$          | the potential level, $E_g$ (1,000 in <code>rule.json</code> ; $10^6$ in the two slits’ configuration file <code>light.json</code> , gravity off in all but name) | 1 + 3 + 3 real lines, sourced by the form, the three stresses and the field’s count flux                                           | the potential level, its own level among it                                                    | $E_g$ , the hierarchy; its rest $c_{vac}$ , 60 by declaration                                                                               |
| the charge potential, whose quanta are light, born by the write                                                      | $[1, 1]$                              | the sign, 1                                                                                                                                                      | 1 real line, sourced by the Wronskian of its readers; under the rotation 1 + 3, three odd axis lines sourced by the sign’s current | the level’s potential fields, at the weight 1; never its own level                             | $k$ , the write weight ( <code>write_weight</code> : 1 in the rule’s own file; the calibrated value of Section 5.1 in the others), declared |
| the binding potential, not the strong force; its own events have the gap $\cos \omega = \text{num}_b / \text{den}_b$ | $[2400, 2401]$                        | the potential level, 1 ( $10^6$ in <code>light.json</code> )                                                                                                     | 1 real line, sourced by the form                                                                                                   | the potential level, its own level among it                                                    | the pair, its range $R = 20$ Links following from it (S.11)                                                                                 |
| matter, the scalar; a quantum that leaves a bound state is a wave packet of matter                                   | $[m, \text{den}]$<br>$[(4000, 6000)]$ | nothing                                                                                                                                                          | dimension 1, one real line; a charged field of the same pair is a second entry of dimension 2, a plane                             | the level’s potential fields, $x = (L_b + L_g) / \Gamma$ ; the plane also the charge potential | $m$ , the mass                                                                                                                              |

glossary’s sense), this world’s declaration above two floors the law gives, the lattice’s own line’s floor for a kick and the swing read on the lattice of a world with bound states in a closed box (derived for the floor, the swing read on the lattice; lattice; the long version’s Section 6.5 and S.62 at its tag, `paper-long-v1.1`); no click bounds it above, since every potential field reads the level and a uniform rest is a change of units (S.12, S.48) (declaration above two floors; derived for the speeds; lattice for the level and its floors, clicks for the speeds; S.12, S.48). Under the law’s line the massless potential’s own travelling events have light’s paces identically in every level,

so gravity's events and light through the same wells arrive together,  $\gamma_{\text{GW}} = \gamma_{\text{EM}}$  (the gravitational wave's and the electromagnetic wave's), where GW170817 bounds the difference between  $-2.6 \times 10^{-7}$  and  $1.2 \times 10^{-6}$  [37] (a comparison); since the two speeds are one speed, the measurement bounds no rest.

## 4.2 The bound state: the fixed point and its stability

A bound state (the long version's body) is its count and its configuration over its Nodes, a configuration held at its Nodes by the potential wells; its count is the number of its quanta, it clicks when a quantum from outside meets its own transition, and what a NodeDetector over all its Nodes credits of it is the energy of its configuration in quanta, a reading and not a state. No law is written for a bound state: it is the support of a standing mode of the one rule, the fixed point of its binding potential, standing where the sources return themselves, its quanta its configuration's quantum share, and a free packet or a cloud falls as  $a = \Delta |d\omega_0/d\ell| \nabla \ell$ ,  $\Delta = 1/m^*$  the band's curvature at rest and  $\ell$  the potential level (derived from the line; clicks; S.18). The free packet's fall coefficient is  $2\nu/(3(1+\nu))$ ,  $\nu = \text{num/den}$ , differing between fields at finite gaps,  $4/15$  at  $[2, 3]$  against  $2/7$  at  $[3, 4]$  (derived; clicks; S.18, S.44, and S.53 of the long version); a bound state in the dark is only the future and still gravitates. It is computed by the rule's own iteration, the wells a spread of the declared count lays, the configuration's top mode in those paces by Rule3's own read act, the levels scaled to the declared count and the wells rewritten until the levels repeat, in integers, at the fixed point of the division act; the branch is chosen by the initial data's width alone (S.45).

For a bound state of  $C$  quanta with the configuration  $\phi$  over its region,  $\sum \phi_i^2 = 1$ , sourcing potentials with the static kernels  $\mathbf{G}_r$ , the fixed point is to first order in the well a critical point on the sphere of the functional  $F[\phi]$  of S.13, the band's pressure less the wells' energy, each well counted once because the configuration is its own source (Hartree's form), with  $2 \cos \omega_b = 2 \cos \omega_0 - \lambda$  at  $dF/d\phi = 2\lambda\phi$  (a theorem); under the dilation  $\phi_\lambda(x) = \lambda^{-d/2}\phi(x/\lambda)$  the pressure is  $K/\lambda^2$  (a second  $\lambda$ , the dilation's) and each potential's well energy  $V_r/\lambda^{s_r}$ ,  $s_r$  its scaling power at the bound state's size.

**Proposition 1** (the stable bound state, under the dilation and to first order in the well) *For power-law kernels, at a fixed point of the binding potential the virial  $2K = \sum_r \sigma_r s_r V_r$  holds,  $\sigma_r = +1$  for a hollow and  $-1$  for a hill, and the second variation of  $F$  along the dilation is  $F''(1) = \sum_r \sigma_r s_r (2 - s_r) V_r$ . A bound state is stable along the dilation only where this sum is positive; the statement is along the dilation alone, and a full minimum of  $F$  would need the Hessian's mixed block, which no derivation here gives.*

Derived in S.45 by Derrick's scaling [38] and Pohozaev's identity [39] on the lattice, for the continuous functional and to first order in the well, the integer step's version open: a massless potential holds the bound state; a hollow beyond its reach adds a barrier with no critical point where  $3bc > a^2$  ( $a, b, c$  the pressure's, the hollow's and the well's coefficients at unit size), the collapse, which ends in a frozen clock under the one assumption (S.39); and a screened kernel's exponent runs from the massless kernel's 1 to the contact kernel's 3 (S.13). The binding potential's fixed point has two branches,

a wide cloud and a compact pixel, and the initial data's width alone chooses between them (theorem for the equation; derived for the stability, the integer version open; computed for the branches; lattice; S.13, S.45); no standing bound state is claimed from a run (computed, a lattice reading, named; lattice): the computed one is coarse in the lattice's pixel by construction, its drift and breathing the lattice's budget and the initial data's,  $(\text{Link}/R)^2$  and  $\sqrt{n}/A$  (S.53), and whether a laid bound state stands under Rule3 within that walk is a test of the archive and not this paper's claim.

The atom is known from nature's measurements and nothing of it is seen on the lattice; what is tested is whether the way from the clicks to the lattice and back carries it (S.43, S.52). In the model an atom is a bound state of the charge potential: a hydrogen nucleus, one quantum of the proton's field, writes the time angle  $\theta_0 = Z\alpha c/r$ , with  $c = 1/\sqrt{3}$ , into the charge potential by its Wronskian, electrons, quanta of the matter field, read that angle into their rotation, and the standing configurations Rule3 makes in that angle are the atom's levels, with Bohr's form  $E_n \propto 1/n^2$  and  $a_0 = \sqrt{3}/(m^*Z\alpha)$  when the nucleus's angle alone is read (derived in the slow limit under line (1) of S.52, an assumption, fitted; line (2) an assumption; the criterion the model's decision, an assumption; clicks; S.43, S.52); a configuration that reads its own charge field has the law's form at large  $Z$  only,  $1 - 1.25/Z$ , and not hydrogen, so under line (1) of S.52 a configuration reads every charge field but its own, and the level's potential fields keep the self-read (line (2), an assumption). The criterion is a rule of reading: what nature splits with energy is read as a bound state, and what no energy splits as a quantum of a field, a quantum itself never breaking; the atom, the nucleus and the molecule are bound states by the one mechanism the law has, a configuration standing in a potential's well (the model's decision, an assumption; the one mechanism an assumption; S.52 of the long version).

### 4.3 A bound state's clicks

What a NodeDetector reads of a bound state follows from its configuration and the paces, with no further line, and none of it is the result of a run. A steady rotation writes no light, and a uniformly moving bound state writes no travelling light in the long-wavelength band (S.40 (b)): a bound state at rest with the plane configuration  $\text{re} = A \cos \omega_b t$ ,  $\text{im} = eA \sin \omega_b t$ ,  $e$  its ellipticity, the ratio of the two parts' amplitudes, has the Wronskian  $W = -eA^2 \sin \omega_b$  at every interval, so the charge potential's source is constant and there is no click; a breathing moves  $W$  between the Nodes, and the light written by the second moment of the charge carries the frequency of the breathing and not the rotation (S.40). For one Node of a bound state,  $a_{\text{next}} + a_{\text{before}} = 2 \cos \omega(t) a_{\text{now}}$  with  $\omega$  changing slowly keeps  $A^2 \sin \omega_b$  invariant, the discrete adiabatic invariant of Rule3's line, while the form  $D = A^2 \sin^2 \omega_b$  and the energy are not invariant, the recurrence stepped over 20,000 intervals with  $\omega$  lowered from 0.841 to 0.600 keeping  $D/\sin \omega$  within  $4 \times 10^{-5}$  while  $D$  falls by a quarter (computed; S.14). The quantum share at a Node is  $(\Gamma^2/p_i^2) A^2 \sin^2 \omega_b/T$ , and under a moving well it drifts with no current through any Link,  $e_i(t_2)/e_i(t_1) = [\sin \omega_b(t_2)/\sin \omega_b(t_1)] [p_i^2(t_1)/p_i^2(t_2)]$  (S.14).

The same identity is Planck's relation on the lattice:

$$\boxed{\text{quantum share} = I \sin \omega, \quad I = A^2 \sin \omega / T}, \quad (7)$$

so  $E = \hbar \omega$  holds, with  $T$  for  $\hbar$  and  $\sin \omega$  in place of  $\omega$ , the invariant standing where nature's number of quanta stands and the energy where nature's energy stands (derived; clicks; S.14). A bound state's rotation reads the level once, so at rest its rate falls by  $f(\omega_0)U$  with  $f(\omega_0) = 2(1 - \cos \omega_0)/(\omega_0 \sin \omega_0)$ , 1.063 at [2, 3] and 1 as the gap closes,  $f = 1 + \omega_0^2/12 + \dots$  in series, the finite-rotation correction in  $\omega$  of a rate that is exactly  $N$  in  $2 \sin(\omega/2)$  (S.15, S.31); a light clock between two faces falls by  $2U$ , the Link's pace read twice. A bound configuration at the rotation  $\omega_b$  that reads a light level is modulated by  $\epsilon = 4g(1 - 2x_0)b_L/\Gamma$ ,  $g = 1 - \text{num}/\text{den}$ , the rest term's coefficient (half the mass term  $m^2$  of Section 2.2),  $x_0$  the bound state's potential level in units of  $\Gamma$  and  $b_L$  the light level's amplitude, the resonant part at  $\omega_L = 2\omega_b$  growing the amplitude at  $\mu = \epsilon/(4 \sin \omega_b)$  per interval, and a bound state with two modes  $\omega_i < \omega_j$  transfers between them at the difference, at Rabi's rate  $\Omega_R = \epsilon_{ij}/(4\sqrt{\sin \omega_i \sin \omega_j})$  (derived, the invariant computed; clicks; the drift lattice; S.14, S.15, S.16, S.41), under the charge potential's rotation in the small-line form  $\Omega_R = \theta_L \omega_L |d|/2$ ,  $\theta_L$  the light's angle on the Link and  $d$  the transition's dipole element (S.41); the fraction absorbed, with the bound state's write back, is open. A bound state's count in clicks is its energy summed over its Nodes, a gapped quantum's energy in a well differing from  $W_{\text{rec}}$  by the rest term's shift  $4U(\text{den} - \text{num})/\text{den}$ , the mass defect read as the count. A NodeDetector with Nodes alone credits the energy, a bolometer's reading, Millikan's slope 0; a NodeDetector with a configuration of its own takes one transition per click, the energy  $T(\sin \omega_j - \sin \omega_i)$ , one photon's  $T \sin \omega_L$  to the order  $\omega_i \omega_j$ , below  $10^{-18}$  at nature's gap, so that a bound state counts photons and reads  $\hbar \omega$  per click, Millikan's slope  $T$  (S.14). At the atom the born light is at the resonance  $\omega_e - \omega_g$  and not at the energy-conserving frequency, absorption erases the photon's empty wave inside the click's causal cone, there is no recoil in the law's write, since the write conserves no momentum (under the proposition of Section 2.5.2 the twist is  $\text{sgn}(k)(|k| \text{div } M)$  per Link and the recoil  $k^2/(2Mm^*)$  in rotation; the atom's needs the nucleus's finite inertia, which line (2) of S.52 pins, so it stays outside the claim), and the orbital Zeeman splitting has Landé's  $g_L = 1$  (S.43). Two standing bound states at the two ends of a Link write a cross current at their beat, which falls under one quantum's measure beyond the reach  $d = \ln(\text{num } a_1 a_2 / (3 \text{den } T \sin \omega_0))/\kappa$  (derived; lattice; S.17).

#### 4.4 Clusters: the three kinds of binding, and the cosmology at small redshift

The model writes no line for a cluster: every bound state is bound and configurations may be free, a free configuration gravitating by its form, falling as every field falls and clicking only where it meets a declared NodeDetector's transition, and a cluster is bound states standing in one another's wells. The three potential levels at a Node are the nesting, the binding potential's level the matter's tail, the charge potential's the charge's well and the massless potential's gravity's well, three kinds of binding by rank, quanta into a bound state, bound states into a charged cluster, charged

clusters into a gravitational cluster (derived from the reads of Section 4.1; lattice); the couplings are at the edge, every divisor 1 but gravity's  $E_g$ . A bound state declared a NodeDetector clicks in the meeting's form, and one with no declaration does the Nodes' laws alone; a click is not the NodeDetector's privilege, every bound state, in the model's intent and in the hypothesis of Section 9, absorbing, emitting and converting by the same meeting, the NodeDetector being the one bound state whose clicks the run reports, and what it credits of a bound state moves only by the clicks at its boundary, up to the drift of Section 4.3. What nature defines as dark matter, gravitating and unseen, is in Universe24 every configuration that meets no NodeDetector's transition, the free quanta of the gapped scalar matter field and the bound states no declared NodeDetector meets, writing the potential level by its form, clicking nowhere and scattering no light (definition, derived under the reads of Section 4.1; the source a hypothesis; lattice, clicks); its kind is derived under the universe's table as declared, its source is a hypothesis, its amount is not derived, and the paper claims no identification with nature's dark matter. The universe of the model is a finite ball of bound states flying through the static lattice, every one pulled by the inverse square of the others at Kepler's  $G_K$  (Section 5.1): a shell of dust at the radius  $a$  feels the mass inside it, so  $\ddot{a} = -(4\pi G_K/3)\rho a$ ,  $\rho$  the dust's density, and  $\dot{a}^2/a^2 = (8\pi G_K/3)\rho - K/a^2$ , Friedmann's dust equations exactly for the ball's interior [40][41],  $K$  the initial data's declaration, and a receding source's redshift is  $1+z = 1+\beta$  at first order, Doppler's and Hubble's known forms, the two derivations agreeing exactly (derived, two derivations, under the declared homologous start; clicks; S.29). The paper's cosmology is claimed at  $z \ll 1$  and not beyond: at the second order the ball decelerates, bound or free,  $q_0 = \Omega_m/2 > 0$  ( $q_0$  the deceleration parameter,  $\Omega_m$  the matter density parameter), where the supernovae read  $q_0 \approx -0.5$ , a miss named in Section 8.

## 5 The formulas: the ones shared with nature and the ones the paper adds

Each form below, the vector sector's among them, follows from the rule's coefficients, read by the geometric optics of a packet's band; the form of nature beside it is never an input, except for the inputs Section 8.2 names (the one assumption, line (1) of S.52, the counting rule, the click's twist). The coefficients contain no metric, no Lorentz transformation, no potential and no force, only the paces and the Port's phase: the clock composes by the rule, the Link is the clock twice by the one assumption, and the Port's phase is the potential field's odd line, sourced by the count's flux under the same assumption, with no number put in. For a moving source the potential's flux writes three odd Link levels  $V_{ij}$ , the lattice's  $g_{0i}$ , read as a phase by plane configurations and by no real one.  $U = \ell/\Gamma$  denotes the model's potential at the point named, the clock's share per level, which no NodeDetector reads, and  $U_b$  its value at the impact parameter  $b$ .

## 5.1 The formulas shared with nature

A bound state is seen as the model's NodeDetectors see it, by the clicks of the light its parts write and absorb, redshifted by the clock's share and, under the one assumption, bent by  $4U_b$  on the way; the clicks' pattern over time is the fall (Sections 4.2 and 4.3; this section). The forms are stated in two potentials. Every test of gravity takes  $GM$  from Kepler's law of the orbits of matter and  $c$  from light, so it reads  $U_K = G_K M / (c^2 r)$ ; in the model a free packet of matter of the pair [num, den] falls in a potential gradient at  $a = c^2 [2 \cos \omega_0 / (1 + \cos \omega_0)] |\nabla U|$ , to first order in  $U$  and at small wave number, the coefficient exact in the gap (derived; clicks; the index lattice; S.18, S.21, S.22, S.28, S.33, S.44), so the orbits of matter measure

$$U_K = \frac{2 \cos \omega_0}{1 + \cos \omega_0} U, \quad 0.800 \text{ at } [2, 3], \quad 1 \text{ as the gap closes.} \quad (8)$$

In  $U$  the composed paces' metric gives, in the post-Newtonian words,  $\gamma = 1$  and  $\beta = 1$  exactly, the metric  $N^2 = e^{-2U}$  and  $h^2 = e^{2U}$ , and the redshift's factor  $f(\omega_0) = 1 + \alpha_U$  with  $\alpha_U = 0.063$  at  $[2, 3]$ ; in  $U_K$ , for  $0 < \text{num} \leq \text{den}$ , the same algebra gives  $\gamma_K = \text{den}/\text{num}$ ,  $\beta_K = (\text{den} + \text{num})/(2\text{num})$  and  $1 + \alpha_K = \tan \omega_0 / \omega_0$ , that is 1.5, 1.25 (the small-gap closed form under the quadratic band; the lattice band's advance is in the list of Section 5.2) and 1.329 at  $[2, 3]$  and 1 as the gap closes, the computed pair's values the lattice's illustration of the band at a coarse gap and nature's rows standing beside the formulas at nature's pair alone (derived under the one assumption; the bending's measurement its calibration, the perihelion the test of the end's choice, the rest its consequences, the delay and the redshift read against Cassini and Pound and Rebka, the shadow bounded by measurement and not confirmed, conditional on the interior's fate; clicks; the computed pair's values lattice; S.18, S.19, S.20, S.30, S.31, S.51). The bending of light is the case  $\beta = 1$  of the push on a moving configuration, light's own speed,  $\beta$  here the speed ratio of Eq. (9) and not the post-Newtonian  $\beta$ ,  $\alpha$  the bending angle,

$$\alpha = \left| \int \frac{\partial n}{\partial b} dl \right| = 4U_b, \quad \mathbf{F} = \frac{E^2 + P^2}{E} \nabla U = E(1 + \beta^2) \nabla U, \quad \beta = \frac{P}{E} = \frac{v}{c_{\text{local}}} \quad (9)$$

at the small gap, with  $E = \omega$ ,  $P = cN^2 k$  the momentum at the composed paces,  $\beta$  the configuration's speed over the local light speed  $cN^2$  and  $U_b = 3s/(4\pi\Gamma b)$ ,  $s$  the bound state's write per interval over the divisor  $E_g$  (S.20, S.51); at rest the push is  $E \nabla U$ , for light  $2E \nabla U$ , the factor two being the ruler going with the clock, and at the computed pair the coefficient of  $\beta^2$  read against light's speed is  $\gamma_K - \alpha_K/2 = 1.335$ , against the band's kinetic scale, where the same push has the coefficient 1.0045 (derived under the one assumption; clicks; S.51). A stationary boundary conserves the rotation, so for every field a configuration of wave number  $k$  entering a potential level  $U$  takes the

wave number  $k_{\text{in}}$  given by

$$1 - \cos k_{\text{in}} = \frac{1 - \cos k + 3(\text{den} - \text{num})(1 - N^2)/\text{num}}{N^4}, \quad N = e^{-U}, \quad (10)$$

light's index  $k_{\text{in}}/k \rightarrow e^{2U}$  at long wavelength (the optical-mechanical analogy's index), for light exact as  $\sin(k_{\text{in}}/2) = \sin(k/2)/N^2$  with the lattice's chromatic term  $k^2(e^{6U} - e^{2U})/24$  at finite  $k$ , and slow matter's index  $n_m^2 = [1 + 6(\text{den} - \text{num})(1 - N^2)/(\text{num} k^2)]/N^4$  (S.28): a slab of potential bends every configuration toward its thick side, the fall of a bound state is the refraction of its own configuration, and a slab thick in the middle is a lens whose image is built click by click. In Universe24 it computes, the quantum of action is one click, whose action is the quantum's norm  $T$  and whose energy is  $T \sin \omega$ ,  $T$  at  $\omega = \pi/2$ , the quantum share's ceiling; a bound state whose write is  $s$  quanta per interval over the divisor  $E_g$  makes the rest of the massless potential  $3G_0(r)s$  at  $r$  Links,  $3s/(4\pi r)$  far away, and the level slows the clock by Newton's potential  $\Phi = -c^2 U = -c^2 \ell/\Gamma$ , with Newton's sign, so (S.25)

$$\frac{G_{\text{clock}}}{c^2} = \frac{3}{4\pi \Gamma E_g}, \text{ that is } G_{\text{clock}} = \frac{1}{4\pi \Gamma E_g}, \quad (11)$$

per quantum of well, in Links and intervals, a reading of  $\Gamma$ , the file's  $E_g$  and the lattice; Kepler's  $G$ , read from the orbits of matter as clicks, is  $[2 \cos \omega_0/(1 + \cos \omega_0)] G_{\text{clock}}$  by Eq. (8), 0.8 of it at [2, 3]. Under the charge potential's rotation the quantum of a field is one unit of the invariant,  $2A^2 \sin \omega_s = T$ , and its Wronskian  $A^2 \sin \omega_s = T/2$  for every field, the unit of charge; the law's coupling between two quanta,  $\alpha_{\text{law}}$  (the item of Section 5.2), carries  $k$ , the charge potential's declared write weight (not the wave number), and nature's  $\alpha = 1/137.036$  [42] fixes  $k/(\Gamma E_s)$  and nothing else, so  $\alpha$  is a coefficient of the file, read from one click as  $G$  is (derived;  $k$  declared; charge universality and the forces' ratio in the small-gap limit, by construction, no test; clicks; S.24, S.25, S.26, S.27, S.42); the force on a bound state of  $C$  quanta is  $\pm C \nabla \theta$ , Coulomb's  $q_1 q_2$  with  $q = \pm Ce$ . The two potentials are one kind of field with one kernel, so gravity's coupling between two quanta of the field  $s$  is  $\alpha_g(s) = (3\sqrt{3}/4\pi) \sin^2 \omega_s/(\Gamma E_g)$ , and between two fields the weights cancel,  $(F_e/F_g)_e/(F_e/F_g)_p = (m_p/m_e)^2$  with no number, by construction and no test: nature's  $4.1656 \times 10^{42}/1.2356 \times 10^{36} = 3.37 \times 10^6$  against  $1836.15^2 = 3.37 \times 10^6$  [42], an identity in the small-gap limit forced by charge universality and  $F_g \propto m^2$  (S.42). The units are the electron's, one interval  $\omega_e \hbar/(m_e c^2) = 1.288 \times 10^{-21} \text{ s} \times \omega_e$  [42] and one Link  $\sqrt{3} \omega_e \bar{\lambda}_e = 6.69 \times 10^{-13} \text{ m} \times \omega_e$  ( $\bar{\lambda}_e$  the electron's reduced Compton wavelength); what a click reads of a quantum is  $T \sin \omega$ , at most  $T$ , against the electron's rest energy  $T \sin \omega_e$ , so LHAASO's 1.4 PeV photon of 2021 [43] bounds  $\gamma_K - 1 = \omega_e^2/2$  below  $6.7 \times 10^{-20}$ , and the Link's bound from MAGIC's subluminal dispersion limit [44] gives  $\gamma_K - 1 < 2.0 \times 10^{-28}$  along an axis, under the premise on the mean that S.27 states, the steps through  $\omega_e$  and den in S.24 and S.27 (derived form, bounded by LHAASO and MAGIC and confirmed by neither; clicks; S.24, S.27).  $\Gamma$  reads nothing of  $\omega_e$  or of  $\alpha$ , and no click of the clocks bounds it: the clocks read the mean of the write's dithered sub-unit level, the exact

rest, and not  $\Gamma$  (derived, an argument on the mean; the dynamics under a dithered level are open; lattice; S.26); the gap bound binds the matter field's own denominator. *The formulas shared with nature, each reached from Rule3 in lattice form, with nature's form beside it, its status and fence (the key of Section 1.3) and its derivation.  $U$  is the clock's share and  $U_K$  the Newtonian potential the orbits measure.*

- The lattice Klein-Gordon equation,  $c_K^2 = \text{num}/(3\text{den})$ ,  $m^2 = 2(1 - \text{num}/\text{den})$ ; Schrödinger's slow limit with  $m^* = 3 \tan \omega_0$ ; nature's form Klein-Gordon; Schrödinger (derived; lattice; S.1, S.34 (c), S.53).
- $E = m^* c_m^2 = \omega_0$ ,  $c_m^2 = \omega_0/(3 \tan \omega_0)$ ; nature's form  $E = mc^2$  (derived for  $m^*$  from the band,  $c_m$  defined, the identity by construction and no test; lattice; S.18, S.21).
- The conserved form (Theorem 3) and the count as the quantum share (Theorem 4); the Wronskian as the charge; nature's form the conservation of energy and charge (theorem at static paces, for the line without its division; lattice; S.5, S.6, S.9, S.45).
- Poisson's rest  $3s/(4\pi r)$  of the massless potential; the reach  $R = \sqrt{\text{num}/(6(\text{den} - \text{num}))}$  of a gapped one; nature's form Poisson; Yukawa (derived; lattice; S.11, S.37).
- The adiabatic invariant  $A^2 \sin \omega$  and Planck's relation, Eq. (7),  $T$  for  $\hbar$ ,  $\sin \omega$  for  $\omega$ ; nature's form  $E = \hbar \omega$  (derived; clicks; S.14).
- The clock's rate  $1 - f(\omega_0)U$ ,  $f = 2(1 - \cos \omega_0)/(\omega_0 \sin \omega_0)$ , 1.063 at [2,3]; a cavity of bound states and an atom's beat shift alike by  $N$ ; nature's form the gravitational redshift, Pound and Rebka [45];  $z = (1 + \alpha_K) U_K$  with  $|\alpha| < 2 \times 10^{-4}$  in nature [46] (derived under the one assumption; clicks; S.15, S.31).
- The bending  $4U_b$ , Eq. (9); against  $U_K$ ,  $(1 + \gamma_K) 2G_K M/(c^2 b)$ ; nature's form Einstein's  $4GM/(c^2 b)$ ; Cassini's  $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$  [47] (derived under the one assumption, which this click fixed (the calibration, no test); clicks; S.20, S.51).
- The delay  $2\sqrt{3} \mu \ln(4x_1 x_2/b^2)$  intervals at  $\gamma = 1$ ,  $\mu = GM/c^2$  in Links,  $x_1$  and  $x_2$  the signal's two ends' distances; nature's form Shapiro's delay [48] (derived under the one assumption; clicks; S.20).
- The perihelion  $6\pi\mu/(a(1 - e^2))$  per orbit at the small gap; against  $U_K$ ,  $(2 - \beta_K + 2\gamma_K)/3$  of Einstein's, the lattice band adding its  $k^4$  term at a finite gap (the list of Section 5.2, S.19); nature's form Einstein's advance, the click that fixed the assumption, Mercury's 43 arcseconds per century;  $|\beta - 1| < 10^{-4}$  [46] (derived at the small gap under the one assumption, fixed by the bending's measurement, the perihelion choosing the reading and testing the choice, a test of the choice, one bit, and no test of  $\beta$ 's value; the quadratic band an assumption at the computed pair; clicks; S.19, S.34).
- Frame dragging: a plane configuration's frequency shifts by  $\delta\omega/\omega_0 = -4\mathbf{V} \cdot \mathbf{v}$  exactly under the Port's phase, with  $\mathbf{V}$  the odd lines' level, the post-Newtonian vector potential of S.56, Einstein's  $-4\mathbf{V} \cdot \mathbf{p}$  for a slow mass; nature's form Lense and Thirring's, the gyroscope's precession 39.2 milliarcseconds per year in the harmonic form, Gravity Probe B's  $37.2 \pm 7.2$  [49] (derived under the one assumption, at nature's gap; clicks; S.35 (c), S.50, S.56 (g)).
- The preferred-frame parameters  $\alpha_1 = 0$  and  $\alpha_2 = 0$ , the harmonic form  $-4V_j$  the standard post-Newtonian form under a relabelling of the reading; nature's  $|\alpha_1| < 4 \times 10^{-5}$  [50],  $|\alpha_2| < 4 \times 10^{-7}$  [51] and  $< 1.6 \times 10^{-9}$  [52] (derived under the one assumption, at nature's gap; clicks; S.35, S.50, S.56 (g)).
- The Nordtvedt parameter  $\eta_N = 4\beta - \gamma - 3 - \alpha_1 + 2\alpha_2/3 = 0$  at nature's gap, 0.5 at the computed pair ( $\beta_K = 1.25$ ,  $\gamma_K = 1.5$ ); nature's  $|\eta_N| < 10^{-3}$  from lunar laser ranging [46] (derived under the one assumption, at nature's gap; clicks; S.56 (g)).
- The moving clock  $f_v = 1 - v^2/(2c_m^2) - (2\omega_0/\sin 2\omega_0) v^4/(8c_m^4) + O(v^6)$ , Lorentz's at  $c_m$  to the second order;  $c_m/c = 0.867$  at [2,3]; nature's form Lorentz's factor; the limiting speeds of matter and light equal: the subluminal case at  $6 \times 10^{-20}$  [53], the superluminal at  $5 \times 10^{-21}$  [54], the earlier bounds at  $10^{-15}$  from photon decay to  $10^{-23}$  for the proton [55] (derived; clicks; S.21).
- De Broglie's  $k = \omega_0 v/c_m^2$  to second order; nature's form  $\lambda = h/p$  (derived; clicks; S.22).
- The fall  $a = \Delta |d\omega_0/d\ell| \nabla \ell$ , universal within a field; between fields  $\eta = 32.3 \omega_e^2$ , a free-packet upper estimate, below  $1.3 \times 10^{-26}$  at the MAGIC bound and  $4.4 \times 10^{-18}$  at LHAASO's; nature's form Galileo's fall; Eötvös;  $10^{-15}$  for titanium against platinum [56], the model's  $138 \omega_e^2$  for titanium against platinum at natural abundances (135.6 for the single isotopes  $^{48}\text{Ti}$  and  $^{195}\text{Pt}$ ),  $1.9 \times 10^{-17}$  at the bound (derived; clicks; S.18, S.24, S.44).
- Newton's pull  $1/r^2$  and  $G$  as a reading, Eq. (11); Kepler's  $G$  at 0.8 of the clock's at [2,3]; nature's form Newton, Kepler (derived; lattice; clicks; S.25).
- The lens, Eq. (10), one formula for light and matter; nature's form  $n = 1 - 2\Phi/c^2$ ;  $n^2 = 1 - 2\Phi/v^2$  (derived, two derivations; lattice, clicks; S.28).
- Coulomb's force over  $\gamma_L$  (the Lorentz factor) between moving charges, 0.995 at 0.1c and 0.866 at 0.5c; magnetism as the Link's phase; nature's form Coulomb, Lorentz, Peierls' coupling (theorem for the conservation, derived for the force's form under the rotation, not under the plain read; lattice for the identity, clicks for the force; S.33).
- $\alpha = (3\sqrt{3}/8\pi) k/(\Gamma E_s)$ ,  $k$  declared; charge universality  $|q_p| = |q_e|$  by construction; the two forces' ratio  $(m_p/m_e)^2$  between fields; nature's form  $\alpha = 1/137.036$ ; by construction and no test,  $3.37 \times 10^6$  against  $1836.15^2 = 3.37 \times 10^6$  (derived,  $k$  declared; no test; clicks; S.26, S.42). Nature's equality holds to  $10^{-21}$  [57] (a comparison).
- Rabi's rate  $\Omega_R = \epsilon_{ij}/(4\sqrt{\sin \omega_i \sin \omega_j})$ ; the parametric line at  $2\omega_b$  under the pace read alone and none under the rotation (S.41); nature's form Rabi (derived, two derivations (S.16); clicks; S.16, S.41).

- A steady rotation writes no light; a breathing bound state writes at its beat; a bound state counts photons at Millikan's slope  $T$ ; nature's form Larmor's silence of a static charge; Millikan (derived; clicks; S.40, S.14).
- Bohr's levels  $E_n \propto 1/n^2$ ,  $a_0 = \sqrt{3}/(m^* Z\alpha)$ , with the nucleus's angle alone; Landé's  $g_L = 1$ ; nature's form Bohr; the orbital Zeeman effect (derived in the slow limit; an assumption, fitted, for line (1) of S.52; clicks; S.43).
- Hubble's law  $1+z = 1+\beta$  and Friedmann's dust equations at  $z \ll 1$ ; nature's form Hubble; Friedmann (derived, two derivations; clicks; S.29).
- The two slits' fringe spacing  $\lambda L/d = 16$  Links and the near-field minima at  $y = 15.3$  and  $32.7$ ; nature's form Young's fringes (derived; clicks; S.23); the counts and the run in Section 6.
- Bell's  $E = \cos 2(a-b)$ ,  $S = 2\sqrt{2}$ , Tsirelson's  $2\sqrt{1+\rho^2}$ , GHZ's  $M = -4$ ; every local counting rule gives at most 2; nature's form Bell, Clauser-Horne-Shimony-Holt (CHSH), Tsirelson, GHZ [58] (the counting rule declared and fitted, the rest derived under it; clicks; S.10, S.38).
- $\gamma_{\text{GW}} = \gamma_{\text{EM}}$  (the gravitational wave's and the electromagnetic wave's), gravity's events at light's speed in every level (derived; clicks; S.12). Nature's form is GW170817's bound between  $-2.6 \times 10^{-7}$  and  $1.2 \times 10^{-6}$  [37] (a comparison).

## 5.2 The formulas the paper adds, and the criterion of refutation

In the model's own potential the bending and the delay are Einstein's to the first post-Newtonian order for a static source and the perihelion is Einstein's at the small gap; the post-Newtonian column at the computed pair departs with the gap, so at the rule's own matter pair the model does not describe the gravity of the solar system, and the closing gap is nature's pair, where  $\gamma_K - 1$  is below the MAGIC bound of Section 5.1 and  $6.7 \times 10^{-20}$  at LHAASO's, whether the model's bound states stand at that pair not shown. Read as a metric, the composed clock gives  $N^2 = e^{-2U}$  and the ruler  $h^2 = e^{2U}$ , against the isotropic Schwarzschild chart's  $1 - 2U + 2U^2$  and  $1 + 2U + 1.5U^2$ : the spatial second order is 2 against 1.5, a prediction no measurement has read, which one click in the deep level would read (S.34). Light's index  $e^{2U}$  places the photon sphere at  $r = 2m$  with  $U = m/r$ ,  $m = GM/c^2$  in Links, so the capture radius of a black hole's shadow is  $b = 2em = 5.44m$ ,  $e$  Euler's number, against Schwarzschild's  $3\sqrt{3}m = 5.20m$ , 4.6 percent larger, a conditional geometric result under the one assumption, known for the exponential metric [16][19] and reached here from Rule3, the ray's capture following from  $n(r) = e^{2m/r}$  alone while the fate of the rays inside is open, the law having no horizon, so that a shadow needs absorption by a bound state's clicks, not computed (S.30); the Event Horizon Telescope's Sgr A\* has the fractional deviation from Schwarzschild  $-0.08 \pm 0.09$  [59], 1.4 standard deviations from the prediction under the Very Large Telescope Interferometer mass prior, and M87\*'s is  $-0.01 \pm 0.17$ , about 0.3 standard deviations [60]. One number follows from the law's lines where general relativity and quantum theory do not meet, entered as untested: in the model a light packet's integrated write is a static bound state's of the same count,  $1 \times E/c^2$  under the static kernel, the packet's own field open, where general relativity's pencil pulls at  $2 \times E/c^2$  [61] (S.39); and one check stands beside it, a bound state laid as one configuration on two paths at two heights having its beat scale as  $N = 1 - U$ , so that the visibility of the fringes falls as  $|\cos(\Omega \Delta U \tau_{\text{hold}}/2)|$ , 0.989 for a strontium clock at one metre for one second, the magnitude of the semiclassical expectation [62], unmeasured (derived under the static kernel, an assumption named, the packet's field open; untested; clicks; S.39). None of the added formulas is confirmed by nature, each following from the line and the assumptions named beside it, bounded

where a measurement exists and a prediction where none does. The criterion is the one stated in Section 1.1: a derived, click-fenced formula that disagrees with a measurement beyond its stated order refutes the model at that formula, and the refutations found are the misses of Section 8.

*The formulas the paper adds, each with its number at the computed pair [2, 3] or at nature's, its status and fence, and the click that would read it; "untested" names a prediction no measurement has read.*

- The post-Newtonian parameters with the gap, for  $0 < \text{num} \leq \text{den}$ ,  $\gamma_K = \text{den}/\text{num}$ ,  $\beta_K = (\text{den} + \text{num})/(2\text{num})$ ,  $1 + \alpha_K = \tan \omega_0/\omega_0$ ; at a finite gap the band's  $k^4$  term adds 13 to 15 percent to the advance, vanishing as  $\tan^2 \omega_0$ , and at nature's gap the closed form is exact (S.19); its number 1.5, 1.25, 1.329 at [2, 3]; 1 with the gap (derived under the one assumption; clicks; the click that reads it: Cassini's  $\gamma_K - 1 = (2.1 \pm 2.3) \times 10^{-5}$  bounds  $\text{den}/\text{num} - 1$ ).
- The click's identity under a write (the long version's booking identity),  $\Delta(wQ) = w(x - v)(x - b)$ ,  $Q$  the step's two-level form of now and before (S.46),  $x$  the written level,  $v$  the level Rule3 would write at the Node and  $b$  the level the step leaves, the quantum share's change under any write (S.46); its number exact at the Node at any paces, with the configuration's own quantum  $W_{\text{rec}}$  (derived; clicks; the click that reads it: every click's write).
- The exponential metric reached from the composed paces,  $N^2 = e^{-2U}$ ,  $h^2 = e^{2U}$ ,  $\gamma = \beta = 1$  in  $U$ ; its spatial second order 2 against the Schwarzschild chart's 1.5 (S.34); its number exact in  $U$ ;  $\alpha_U = 0.063$ , the redshift's factor (derived under the one assumption; lattice, the second order clicks; S.51 for  $\alpha_U$ ; the click that reads it: one click in the deep level).
- The shadow's capture radius  $b = 2em$  against Schwarzschild's  $3\sqrt{3}m$  (S.30); its number  $5.44m$  against  $5.20m$ , 4.6 percent (derived under the one assumption, bounded by measurement and not confirmed; clicks; the click that reads it: the shadow's diameter (Sgr A\* 1.4, M87\* 0.3 standard deviations)).
- The gravity of light, a light packet's integrated write at  $1 \times E/c^2$  under the static kernel, against general relativity's  $2 \times E/c^2$  (S.39); its number the factor 1 (derived under the static kernel (an assumption named), the packet's field open; untested; clicks; the click that reads it: a static bound state's fall toward a light beam (Tolman's pencil), unmeasured).
- Light undragged: a one-part configuration carries no Port phase, so the Lense-Thirring deflection and the Lense-Thirring Sagnac term of light are absent, where every measured drag is of matter; the readings that would read it: a ring laser fixed on the rotating Earth, whose Sagnac frequency carries the Lense-Thirring term in Einstein's metric and none here, the term GINGER is built to read [63], and a spinning black hole's shadow, asymmetric under Kerr's dragging of light and here symmetric under the spin's reversal whatever the spin, where Kerr's shadow departs from a circle by up to about a tenth at the highest spins, a reading of the Event Horizon Telescope's asymmetry [59] (derived under the one assumption; untested; lattice; S.56 (h)).
- The coupling at speed: the Port's angle is the field's  $\omega_0$  times the delay, so a fast configuration's drag falls as  $1/\gamma_L$ ,  $\gamma_L$  the Lorentz factor, against Einstein's  $-4\mathbf{V} \cdot \mathbf{p}$  at every speed, where every measured drag is slow; on a ring the two senses of a matter wave split by  $2v_g\theta = 8v_gV\omega_0$  per Link, and the phase around a closed loop is the circulation of the odd lines' level times  $4\omega_0$ , independent of  $v_g$ , the matter wave's gravitomagnetic Sagnac phase proper (S.56 (e)), the atom interferometers its readers (derived under the one assumption; untested; lattice; S.56 (h)).
- The lens, Eq. (10), with the chromatic term  $k^2(e^{6U} - e^{2U})/24$  (S.28); its number light's index  $e^{2U}$  and slow matter's  $n_m^2 \propto 1/k^2$  (derived, two derivations; lattice for the index, clicks for the focus; the click that reads it: a chromatic bending of light at finite  $k$ ).
- The fine-structure constant's form,  $\alpha = (3\sqrt{3}/8\pi)k/(\Gamma E_s)$ ,  $k$  declared (S.26); its number  $k/(\Gamma E_s) = 1/28.33$  at nature's  $\alpha$  (derived form, the weight declared; clicks; the click that reads it: the one click that fixes  $k$ ).
- The moving clock's fourth order,  $\omega_0 [(\sum_a n_a^4) \tan \omega_0 + \cot \omega_0]$ , anisotropic by  $\sum_a n_a^4$  (S.21); its number 1.6926 along an axis, 1.2224 along a face diagonal, 1.0657 along the body diagonal at [2, 3] (derived; clicks; the click that reads it: a clock flown along the lattice's axes against its diagonals).
- Bell's and GHZ's values at rational settings,  $S = 478/169$  and  $M = -4$  (S.10, S.38); its number  $478/169 = 2.82840$  against  $2\sqrt{2} = 2.82843$  (fitted, an evaluation of the declared counting rule, no prediction; clicks; the click that reads it: the counting rule's own check).
- The clusters' deceleration  $q_0 = \Omega_m/2 > 0$  at  $z \ll 1$ , bound or free (S.29); its number the sign (derived at  $z \ll 1$  under the declared homologous start; clicks; the click that reads it: the supernovae's Hubble diagram stands against it, a miss of Section 8).
- Compton's factor at a finite gap,  $\lambda_{\text{out}} - \lambda_{\text{in}} = (2\pi/(\sqrt{3}\omega_0))(\omega_0/\tan \omega_0)(1 - \cos \theta)$  in the non-relativistic regime, under the twist proposition of Section 2.5.2; its number the factor  $\omega_0/\tan \omega_0$ , 1 at nature's gap, by construction and no test there (hypothesis, fitted; derived under it, nature's by construction and no test; lattice at the computed pair; S.55; the click that reads it: Compton's shift at a finite gap, nature's at 1).

## 6 The two slits: the fringes from the band and the clicks from the counting rule, the one run their check

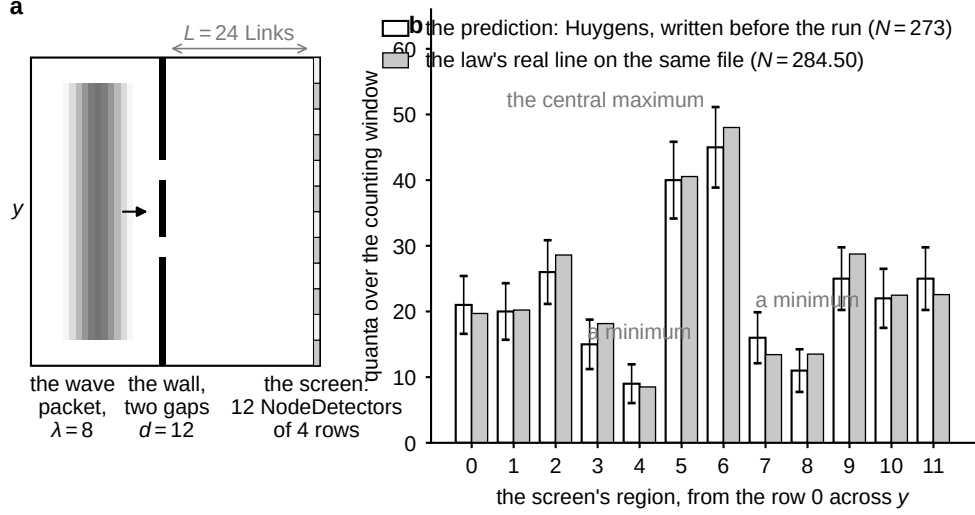
The one run this paper reports is the two slits'. The paper contains no experiment against nature: the two slits is the one computed universe file set against a prediction written before its run, its row read by the one run.

### 6.1 The configuration file

The file is the archive's `two.slits.json` over the fields of `light.json` (Table 1), with its design file `design.json` and its blind file `expectation.json`, run as the reading document `blind.and.reading.md` records: a declared packet of light at the wavelength 8 Links along  $+x$ ,  $k = \pi/4$ , a raised cosine of about ten quanta per Node over the columns 4 to 16, 1,809 quanta by the generator's integer count, 1,810.4 by the exact share's sum (S.23), its before level the band's own exact level mode by mode at the declared vacuum level's paces; the wall with its two gaps of three Nodes, the screen's twelve NodeDetectors of four rows and the counting window as Fig. 2 shows them; a thirteenth NodeDetector, `gap`, with sixteen Nodes alone behind the first slit, backed by no face, whose net inflow over the window is nothing (the real line's  $-0.58$ ), floored at 0 and never drawn; the run 130 intervals of Rule3 at every Node from the file's initial data with the wall's face and the screen's NodeDetectors and nothing else; the flat lattice's folded third axis carrying the file's band exactly at  $k_z = 0$  (computed; lattice; S.23). The file's initial data are the exact initial data, 1,241 levels at the packet's peak and 1,809 quanta, after the design's 1,328 and 1,857 (the law's record [28], the design's lattice); the engine's runs at the level weight 1,000 having shown a self-focusing of light that nature never shows, removed by the light-only file's gravity weight  $10^6$  (fitted, the file's history; lattice; the law's record [28]). On this file the law's real line, stepped by the paper's script `two.slits.real.line.py` (its output `two.slits.real.line.txt` in the paper's folder, at the file of the tag), gives 284.7 units of  $W_c$  through the screen in the physical labels (284.5 in the implementation's labels, the click at the interval  $t$  reporting the pair after  $t - 1$  steps; 278.9 when the laid configuration's own integers, laid at the level 60's paces, are stepped by this line at the level 0's). The peak of the screen's inflow is at the interval 69.0, against 62.0 on the same file with the lattice refined fourfold (62.1 eightfold) (computed; S.23).

### 6.2 What the model says it must give, and the run

The file is read by the law's real line twice, before the run on the design's lattice at the design's initial data, the blind row, and after it on the frozen file at its own initial data; Huygens's construction and Young's spacing applied to the declared geometry give the fringe's spacing and the minima lines (S.23); the run is set beside them. The spacing of the far field is  $\lambda L/d = 16$  Links; at  $L = 2d$  the fringe is the near field's, with the minima near the lines  $y = 15.3$  and  $32.7$ , in the regions 3 and 8 of the twelve four-row blocks counted from 0 (derived; S.23). Light's line at the vacuum's paces, stepped in real arithmetic on the declared file at its own initial data, gives 284.7 units of  $W_c$



**Fig. 2** The two slits, the one computed file (computed; the run's  $N$  a run; clicks). (a) Universe24 as its file declares it. The flat lattice has  $45 \times 48$  Nodes. The wave packet of light, at the wavelength 8 Links, is laid over the columns 4 to 16 (the grey raised cosine) and moves along  $+x$ . The wall at the column 20 has two gaps of three rows, 12 Links apart. The screen at the column 44, 24 Links from the wall, is twelve declared NodeDetectors of four rows each. (b) The screen's row per region over the counting window of the intervals 45 to 130. The design's blind row, the law's line in real arithmetic at the design's initial data and window, by the record of its computation; the archived file's name for it, the Huygens sum), written before the run, is hollow ( $N = 273$ , its bars summing to 275 by the per-Node rounding, with the draw's scatter  $\sqrt{NP(1-P)}$  per region). Beside it, in grey, is the law's real line stepped on the same file at the initial data as the commit holds them ( $N = 284.50$  in the implementation's labels, 284.7 in the physical ones). The run gives  $N = 285$  at both seeds. The points are the seed-24 clicks per region of the one run, from the archived row the draw's gate holds (computed, a run); the run a check, no run made for the drawing

through the screen, region by region, and the integer step's deviation from the line is bounded exactly per step, below one level per Node per interval, the error per laid mode bounded by  $(n-1)/\cos(\omega/2)+1/\sin\omega+1$ , 135.6 at the carrier at  $n = 130$ , while under the model of independent uniform remainders it walks as  $\sqrt{n/(12(1+\cos\omega_k))}$  per band mode, 2.4 levels at the carrier and 3.0 per Node over the lattice's zone at  $n = 130$ , against  $b = 1,241$ , 0.4 to 0.5 percent of the energy and 1.1 to 1.4 units of  $N$  (derived for the bound, computed for the walk; S.53 (i)). The blind row, written before the integer run, is light's own line run in real arithmetic at the design's initial data (the design's lattice,  $64 \times 48$  Nodes with every face open, 1,857 quanta and the window 45 to 120, the law's record [28] the source of the count, cited by the expectation file's comment, against the frozen file's  $45 \times 48$  with receding  $x$  faces, 1,809 quanta and the window 45 to 130; 273 through the screen; `expectation.json`, the design's row). The design's blind row is the per-Node row of 2026-09-30 written before the run from light's line in real arithmetic on the design's lattice, which the archive's file and the long version name the Huygens sum, a static row of the design file that no script of

the archive recomputes, the line's script binding the initial data to the frozen file's digest (the meeting round's 277.8 at its own lay the law's record [28]); and the frozen file's exact initial data, 1,810.4 quanta by Eq. (5), give 284.7 by the same line, and its integer lay, 1,809 quanta, 278.9 (`two_slits_real_line.py` at 1fe3790a); the difference is the design's lattice, initial data and window against the frozen file's, the exact initial data's own number and no scatter's; Huygens's construction and Young's spacing give the minima lines 15.3 and 32.7 and the spacing 16 Links (derived; S.23), the blind row's dips at the regions 4 and 8 the lattice computation's (computed), the pattern's centre at  $y = 24$  being no mirror of the blocks' 23.5 the reason the two readings can differ by one region.

The run (computed, a run),  $N = 285$  in the region's unit  $W_c$  at both seeds, 24 and 25 (674 photons of energy  $T \sin \omega$  at light's own rotation  $\omega = 0.4366$  on the level-60 paces,  $285/\sin \omega$ , a lattice reading; S.61 of the long version [27], `paper-long-v1.1`; the reading document `blind_and_reading.md` at 1fe3790a), at the level 60 on the file's integers, stands within the walk of the line's 284.7 at the level 0 on its own exact initial data; the line's 278.9 on the file's integers, laid at the level 60's paces and stepped at the level 0's, lies 6.1 below the run and outside the walk, the before level's mismatch (up to 18.80 levels, the data file's line 9) carrying an opposite-branch admixture the receding face loses, the mechanism of the meeting round's 277.8 and no comparator of the run; the run stands 12 units over the blind row, the check of the line and not the claim. The sum of the absolute differences of the seed-24 clicks from the prediction's counts is 28 of 275 (Fig. 2(b)), a comparison and not a statistical test, since the blind row is the line's and not a draw. The run's  $N = 285$  and the line's 284.7 are quoted at the file's scale  $s = 1$ ; the lattice's transmission per laid quantum sits above its continuum limit, and the line's  $N$  moves with the scale, the sequence and its limit the long version's (computed; lattice; the long version's Section 9.2, `paper-long-v1.1`). The arrival is  $t = L'/v_g$  with  $v_g = (\text{num}/(3 \text{ den})) \sin k / \sin \omega = 0.547$  Link per interval and  $L' = 34$  Links the flight from the packet's centre to the screen,  $L'/v_g = 62.2$  beside the refined lattice's 62.0 (derived, computed; S.1, S.23). The model's opaque NodeDetector in one gap loses the fringes and reads one gap's envelope, as nature's absorbing which-way markers do, by the continuity of the one inflow through the two boundaries (derived; clicks; S.46); a which-way measurement without absorption is not computed here.

## 7 Bell and GHZ under the declared counting rule

### 7.1 The configuration file

The entanglement the model computes is one event with two trees: a pair is two configurations from one source Node, each read forward and backward, and the pair's correlation is the partner's configuration read through the same source, re-weighted by what was realised. The pair is laid as the pair field, two real lines of light never summed at a Node, with equal parts; each side's NodeDetector is a region at least half a wavelength across, reporting the signed level sums of the two parts over its region, and it applies the declared setting, a pair of integers  $(p, q)$ , as the coefficients of the analyser's two channels  $+$  and  $-$ ,  $e(+) = (p, q)$  and  $e(-) = (-q, p)$ , orthogonal, so

each side's marginal is one half for any setting. The settings of the Clauser-Horne-Shimony-Holt (CHSH) test are (1, 0), (1, 1), (12, 5) and (5, 12), at 0, 45, 22.62 and 67.38 degrees. The counting rule is the one form of the click: the joint energy pairs the parts by their label across the two sides through the source,

$$J(o_A, o_B) = \sum_{\text{window}} \left| \sum_k c_k e_k(o_A) \alpha_k(o_A) e_k(o_B) \beta_k(o_B) \right|^2, \quad c = (1, 1), \quad (12)$$

with  $\alpha_k$  and  $\beta_k$  the amplitude of the part  $k$  at the port on each side,  $e_k(o)$  the port's integer coefficient on the part  $k$ , and  $c_k$  the parts' laid weights, the window's sum standing outside the square. The joint energy is never the square of a local sum, which one NodeDetector could form alone and which gives  $S$  at most 2. One draw is made per pair with the declared seed, the first side from its marginal and the partner from the conditional; each side's write at the Node of its exchange spreads by Rule3 and changes no number of the row, one click per configuration (an assumption).

## 7.2 What the model says it must give

(1) Rule3 gives each part's amplitudes at the ports; the two parts of one beam, laid equal, are two configurations of one field stepped by the same line along the same path, and they are identical bit for bit (a theorem: one integer map on equal integers). (2) The meeting, with the linearity of the real line that Rule3 follows within its bounded remainder (Theorem 2; the bound per laid mode as Section 6.2 states it for the slits), gives the joint amplitude: the realised click at A's port weights the parts by A's analyser, is read back to the source Node, where the two parts were one event, and forward to B, where B's analyser weights them; the source sums the two branches,  $\sum_k c_k e_k(a) e_k(b) = \cos(a-b)$  for the normalised vectors  $e(a)$  and  $e(b)$  (derived; clicks; S.10). (3) That the energy is the density of the meeting, its square, is Born's rule for a pair [32]; nothing in Rule3 forces it, and everything after it is derived in the algebra (a postulate of the law, adopted with nature's correlations in view; the square Born's declaration: the counting rule was chosen to reach nature's correlations, so Bell's  $2\sqrt{2}$  follows from it and does not test it). From the three, with the pair's common phase  $\theta$  uniform over the pairs, A's + port at  $a$  and B's at  $b$ ,  $E$  the correlation of the two sides' signs and  $S$  the CHSH sum,

$$\begin{aligned} \text{the meeting:} \quad & P(++) = \frac{1}{2} \cos^2(a-b), \\ & E(a, b) = \cos 2(a-b), \quad S = 2\sqrt{2}; \\ \text{a local counting rule} \\ \text{by the energies:} \quad & P(++) = \frac{1}{4} + \frac{1}{8} \cos 2(a-b), \\ & E = \frac{1}{2} \cos 2(a-b), \quad S = \sqrt{2}; \\ \text{a local counting rule} \\ \text{by the sign:} \quad & E = 1 - 4|a-b|/\pi \quad (|a-b| \leq \pi/2), \\ & S = 2, \end{aligned} \quad (13)$$

the second and third lines following by averaging over the pair's phase (derived; S.10). So  $E(a, b) = \cos 2(a-b)$  under the meeting, exact for the declared pairs by

Lagrange's identity of the NodeDetector on equal parts: at the four settings  $E = 119/169, -119/169, 120/169$  and  $120/169$ , and  $S = 478/169 = 2.8284$  exactly, the efficiency 1 by construction and the marginal one half on each side whatever the partner's setting. For unequal parts, with  $r$  the ratio of their products at the ports, the same counting rule gives  $E = \cos 2a \cos 2b + \rho \sin 2a \sin 2b$  with  $\rho = 2r/(1+r^2)$  and  $S = \sqrt{2}(1+\rho)$  at the ideal angles, the maximum over the four settings  $2\sqrt{1+\rho^2}$ , Tsirelson's bound [64] at  $\rho = 1$  and never the algebraic 4; the declared pairs' 478/169 sits below  $2\sqrt{2}$  because the angles are rational. No signal passes,  $P(A+) = (\cos^2 a + r^2 \sin^2 a)/(1+r^2)$  being independent of  $b$  for parts of any amplitude ratio.

Without the declaration the lattice gives the second and third lines of Eq. (13), two local counting rules on the same file,  $S = \sqrt{2}$  by the energies of the ports and the sawtooth's 2 by the sign of the larger port, Bell's bound: a pair is one laid configuration, its phase a shared classical variable inside Bell's hidden variable [65], so every local counting rule of it gives  $|S| \leq 2$  (a theorem) [66], and Bell's value sits in the counting rule, a declaration, no independent evidence that the rule accounts for entanglement. A counting rule by one side's contrast (S.10) reaches the quantum correlation only by discarding pairs [67][68][69], the event-based simulations with a time-window rule the named instance [70][71], which the loophole-free tests of 2015 close for nature [72][73][74]. The same counting rule on four lines gives the three-quantum Greenberger-Horne-Zeilinger (GHZ) correlation [75],  $E_3 = \cos 2(a+b+c)$ , with every pairwise  $E$  equal to 0 and Mermin's  $M = -4$  [76], where every local counting rule gives at most 2 (fitted, the counting rule; derived under it; computed for the numbers; theorem for the local bound; clicks; S.10, S.38, S.47). No entanglement other than the pair's and the entanglement swap's is named as met (S.47); the gate-joined ions and transmons that exceed the law's 2 are the miss of Section 8.1. No settings are chosen by the lattice's own past, so a superdeterministic world [77] is not this one. The three prices of keeping the lattice local: Born's rule is the NodeDetector's declaration; a clicked configuration's empty wave still writes its form into the potentials it sources until the front reaches it; and the meeting through the source Node gives  $S = 2\sqrt{2}$  with nothing signalled, resting on the counting rule, a declaration.

## 8 Limits and what fails

### 8.1 The misses, with their numbers

The findings against nature, each set beside its measurement, the derived click-fenced ones refuting the model at that formula (derived, the comparisons supporting findings; clicks; lattice for helium, positronium and the Link's bound; S.35, S.50, S.54). Twelve stand by name: eight findings against nature, a line of the law existing and its number or structure contradicting a measurement, and four places with no line, the neutron's lifetime, Pauli's exclusion, the loops and the  $\times$  polarisation, where nothing is derived and nothing contradicted and the row names what a line would have to give (the key of Section 1.3; the claims list of Online Resource 1).

The massless potential's travelling events carry the breathing polarisation and the lattice-aligned + and no  $\times$  (derived under the static read; S.50 (1)). GW170814's three-detector test favours pure tensor polarization [78] (a comparison). The write without

the odd Link part gave  $\alpha_1 = -8$ ,  $\alpha_2 = -1$ ,  $\eta_N = 7.3$  and no frame dragging (S.35, S.50, the record); under the law's write of Section 2.3 the four are derived at nature's gap and stand in Section 5.1, and the archived implementation, which does not carry that write, reads none of them (Section 8.2). On Rule3's own clock stands this: among five forms tried, no real, local, explicit, bijective lattice line carries  $g_{0a}$ , which is why the odd part is a phase on a plane configuration and no pace (a theorem for one-part real lines with a  $V$ -proportional odd part, the staggered step's whole-Node shift the one exception, S.56 (d); S.36 of the long version its five instances).

At the second order in  $z$  the model's finite ball of bound states decelerates,  $q_0 = \Omega_m/2 > 0$  (S.29), where the supernovae read  $q_0 \approx -0.5$ , the flat  $\Lambda$ CDM model's (cold dark matter with a cosmological constant)  $\Omega_m/2 - \Omega_\Lambda$ ,  $\Omega_\Lambda$  the dark energy density parameter [79][80][81]: Rule3 has no repulsive sector, every potential level held a hollow, so an accelerating Hubble diagram is a finding against nature by name. The nuclear binding under the declared potentials misses in two directions, the deuteron by a factor 4 above and iron by a factor 6 below, and the size law  $R \propto 1/A$  against  $A^{1/3}$ , the same two absences, the exclusion and the repulsive core (derived under the nuclear hypothesis; S.45 of the long version), a bound state's binding per quantum being bounded by  $0.76 CW/(\Gamma E)$  at one Node,  $W$  the declared write weight and  $E$  the divisor, (derived, a bound under the declared  $W$ , which no click bounds; lattice; S.36); the shells, the magic numbers and the neutron's lifetime are not derived.

The model has no electroweak theory, its weak interaction being the conversion click (the declarations of Section 8.2): every rotation potential couples at  $\alpha$  at the Link, so no Fermi constant  $G_F$  follows, nature's  $m_W$ , the W boson's mass [82], at  $\alpha$  falling short by a factor 4.6, the mixing angle's factor absent (S.50); a three-line configuration read as  $(\nu, e)$  under one  $O(3)$  has no fourth rotation, so no  $Z$  boson, where nature has the neutral current [83] and the  $Z$  [84]; parity is conserved, where nature's weak interaction violates it [85], the two one root, a step that commutes with the cube's reflections; and the law has no spontaneous click, so the conversion has no derived rate and the neutron's  $878.4 \pm 0.5$  s [82] has no source.

The atom: the self-read gives hydrogen's  $1s$  at  $-0.043$  of Rydberg's unit (S.43), which line (1) of S.52 mends; the fine structure under the spin-0 configuration has the Klein-Gordon form  $n/(l + 1/2)$ , Sommerfeld's  $n/k$  [86] with  $k$  read as  $l + 1/2$ , splitting hydrogen's  $2p$  from  $2s$  by 29 GHz where Dirac's  $n/(j + 1/2)$  gives  $2p$ 's splitting 10.9 GHz (S.50); the law holds no amplitude over two positions on the lattice, so the reduced mass, the isotope shifts and the muonic atoms are outside what it derives, the many-body correlation is not reproduced by a product of configurations, helium 1.4 percent short of nature's binding energy and positronium at 43 percent of it (computed by hand from the modes, nothing run, a comparison and no click claimed; lattice; S.52), and Pauli's exclusion has no line. Universe24 is a mean-field theory of one quantum: it computes helium at the Hartree-Fock value,  $-2.8617$  Ha [87], and lacks the correlation energy and the radiative corrections in the amounts Hartree-Fock lacks them in nature; its many-body misses (the reduced mass, positronium's binding, Pauli's two per orbital, the nuclear size law in part) are one absence, the parts' pairing, which the credit reads through the root and the lattice's step does not read, antisymmetric with spin for the exclusion, and its radiative misses a vacuum that does

not fluctuate (derived under the two lines where the numbers are, a comparison for the Hartree-Fock value, no line where the key says so; lattice; S.52).

Nature's mass ratios are not whole numbers ( $m_p/m_e = 1836.153$  [42]), and a cluster's defect is bounded by the law at  $C\omega_0 x_{\text{self}}$ , of the order of  $10^{-22}$  at nature's Links by the law's self-read bound ([28], The strong force, (A3)), so no particle of nature can be read as an aggregate of the electron's quanta; the rule's own universe at [2, 3] is no finding against nature, because nature measures at its own pair. The gaps have their causes in the findings above, and none is the fault of a number: no loops ( $\alpha$ 's running, the Lamb shift,  $g - 2$ ), no spontaneous click (the lifetimes), no vector for the exclusion, and a step that commutes with the cube's reflections (the  $Z$  and parity). Between separately prepared bound states the law's counting rules give at most 2, where gate-joined ions [88] and transmons [89] exceed it (Section 7). The lattice's scale: the massless band's quartic term gives the quadratic Lorentz-violation scale  $E_{\text{QG},2} = \hbar c / (\ell_{\text{Link}} \sqrt{\epsilon_4})$ ,  $\epsilon_4$  the band's quartic coefficient, and the Fermi Large Area Telescope's (Fermi-LAT) bound [90] puts the Link at  $\ell_{\text{Link}} < 6.4 \times 10^{-27}$  m, so that the electron's pair needs  $\text{den} \geq 2^{94}$  (S.54; the implementation's width, a bound that is not physics, in the Statements and Declarations). The click's write conserves no momentum, so the recoil is the NodeDetector's, where nature's counter recoils, as in the Mössbauer line [91] and the Compton shift (its form under the proposition is Section 2.5.2's); the model declares a NodeDetector and derives what the click does once declared, and that nature's bound states jump [92][93] is the method's widest open gap. Neither layer keeps a total energy (a theorem for the step's conservations, an assumption for the clicks'). No standing bound state is claimed from a run (the closed box's swing, the archive's one open physics finding, a lattice reading, in the Statements and Declarations).

## 8.2 What is put in

The model declares and does not derive: a field's entry and the integers  $\Gamma = 6,000$ , a multiple of 6 so that light's and matter's pairs are exact over it, the binding potential's pair [2400, 2401] the one pair not over it,  $T = 2^{15}$ ; a NodeDetector with Nodes alone, its region, the file's counting window and the seed; a NodeDetector with a configuration of its own; the emission's rate constant; the conversion's table and rate, a hypothesis under its own name; the initial data; the receding face; the counting rule's one draw through the source Node, the one declared non-local element; and the erasing front (assumption for every declaration; lattice, clicks; the claims list of the supplement). The archived implementation at the tag carries the law's lines without the write of the odd Link part (Section 2.3); no run of this paper uses it, the two slits' file having gravity off in all but name and Bell's derivation reading no current (a declaration of the archive's state; the Statements and Declarations). Three choices were made with a known result in view, and are inputs and not predictions, beside the file's fitted numbers named below: the one assumption  $\hbar N = 1$ , for a source at rest and for one that moves, the odd write of Section 2.3 (fixed by the bending's measurement and tested by the perihelion's, which chose its end; Section 2.3), line (1) of S.52 (the atom's levels: a configuration reads every charge field but its own), and the counting rule, a click's energy as the square of the meeting, chosen to reach nature's correlations

(Section 7); and one line beyond the law, the click's twist of Section 2.5.2, chosen with the per-click conservation of momentum in view, an input to no formula of the paper, Compton's factor its one lattice term under it, 1 at nature's gap, by construction and no test. The numbers of the file are nature's or declarations, none the rule's: the vacuum level, a declaration; the binding range following from the pair (S.11); gravity's weight  $E_g$  and the charge potential's  $k$ , read from  $G$  and from  $\alpha$ ; and the gaps, read from the masses; no number comes from the lattice itself (calibration, experiment for  $E_g$ ,  $k$ , the gap and the Link's length; hypothesis for any such number; clicks; S.48); that the file's universe is the one the paper tests is a decision and not a theorem (an assumption), the free rest's computability a hypothesis (the long version's Section 6.6). The selective read of the nuclear potentials by the nucleon fields and the energy line of S.26 are declared lines of the law. What is held to be computable and not computed: the bound states at the closing gap, the moving fixed point of a bound state, and the absorbed fraction with the bound state's write. What is not claimed: that nature is a lattice; that any number of the computed pair [2, 3] is nature's; that a comparison with nature is more than a consistency statement at the stated order; or that a formula known in the continuum is new where the paper gives its lattice form.

### 8.3 What a world closer to nature needs

This world differs from nature's along two axes: the box's, a finite box returning the potentials' radiation, a box larger than the run's light-crossing ( $10^{12}$  Nodes, 17,000 intervals at  $1/\sqrt{3}$  Link per interval) behaving as open; and  $\Gamma$ 's, the vacuum level's floor  $(K^2\Gamma/16)^{1/3}$  at large  $\Gamma$  being 8 to 13 levels at  $\Gamma = 6,000$  and  $10^{-19}$  of  $\Gamma$  at nature's  $2^{94}$ , so that the discreteness' marks shrink as  $\Gamma^{-2/3}$  and a world closer to nature is a larger  $\Gamma$  with the one-way face, not a larger count of Nodes; the calculations such a world needs, named and not done, none a claim of this paper, are listed in the long version's Section 10.3 at its tag (`paper-long-v1.1`). Universe24 stays as it is, since it is not nature; the lines the algebra reached beyond it (the stresses read into the clock, a write of the click's momentum, a spinor field) are an option for building another world, which would not be Universe24, and no claim of this paper.

## 9 Conclusion

One line of integer arithmetic, Rule3, with the declarations named beside it and no number of nature put in but the two measurements of the one assumption, the bending's that fixed it and the perihelion's that chose the reading and tests the choice, the atom's line (1) and the correlations the counting rule was chosen for, beside the file's calibrations (Section 8.2), gives in lattice form the forms of the list of Section 5.1, each under the condition its item states, and the two slits' row computed at one file; Born's rule is not recovered but put in, with Bell's  $2\sqrt{2}$  its evaluation. Among those forms the vector sector of gravity is the paper's own result: the one assumption, which clocks a Link twice, applied to a moving source gives the Port's phase with no number put in, and with it frame dragging's form,  $\alpha_1 = \alpha_2 = 0$  and  $\eta_N = 0$  at nature's gap, meeting Gravity Probe B [49], the binary pulsar's bound on  $\alpha_1$  [50], the Sun's and the solitary pulsars' on  $\alpha_2$  [51][52] and lunar laser ranging's on  $\eta_N$  [46] by name,

where the write without the odd Link part gave  $\alpha_1 = -8$  and no drag; its two lattice terms stand as predictions, light undragged by a rotating mass, which a ring laser's Lense-Thirring term [63] and a spinning black hole's shadow [59] can read, and a fast record's coupling falling as  $1/\gamma_L$  (derived under the one assumption; clicks at nature's gap; the two predictions untested; S.56). The contribution is the explicit chain from the rule through the declarations and the derivations to the clicks, every input and every miss named, so that each step can be checked and each reading repeated; and, for each field the lists touch, the weak field, the atom's levels, the two slits and Bell's pairs, the cosmology at small redshift, one handle for that field's own research: the formula in lattice form with its status and its fence, to set beside the field's measurements or to refute at it by the criterion of Section 1.1. The paper has one claim, that Universe24 helps compute things in nature, and none of it is a proof about nature: the model is a wave mechanics of configurations on the Nodes and their meetings, what is put in is named (Section 8.2), the form of a click is claimed and its cause is not, and why nature is so, we do not claim to know.

The conjectures stand in the long version at the tag `paper-long-v1.1`, one line each here (inspiration, hypotheses under their own names in the long version; no number of the paper rests on them; clicks; the long version's Section 10.3 and S.37, S.45, S.53 at its tag): the weak force as the conversion click and the strong force as one three-line configuration read whole; the nuclear potentials; the configuration's parts as nature's particles; the masses as fields with one denominator; dark matter's source; the hypothesis that in nature only a click moves a quantum between fields, with the single atom's lifetime as its test; this paper's own, the click's twist of Section 2.5.2, Compton's factor its lattice term, 1 at nature's gap, no test; the frozen Node as the law's one-Link black hole. The paper gives the direction and not the whole, so that others can take the method, with this set of rules or another, and continue.

## Statements and Declarations

**Funding.** None. **Competing interests.** None. **Ethics approval and consent.** Not applicable. **Data and code availability.** The law's document, the implementation, the scripts of the figures and of the derivations, and the long version of the paper are in the public Git repository <https://github.com/Closer24/Universe24>, archived at Zenodo under the concept DOI 10.5281/zenodo.22738745 [94], the record of this version to be deposited before submission, MIT licence, at the tag `paper-long-v1.1` (the commit 441b2399); the two slits' reading at the seeds 24 and 25 in `examples/events/two_slits/blind_and_reading.md`. The implementation is frozen at that tag, whose engine is the code of the commit 1fe3790a of 2026-10-05 the two slits ran at; it is not up to date with the law, whose clock-family write on the Ports it does not step, and whoever wants may update it. The paper builds on the algebra alone. The implementation runs any world its universe file declares, the fields with their pairs and shapes, the holders with their sources, the NodeDetectors and the lay, and holds no number, no formula and no field's name of its own; it is offered as the method's tool for building other worlds than Universe24, the recipe for a new world in `ENGINE.md` of the archive, and no claim of this paper rests on it beyond the

one run it reports, the two slits (Section 6), the check of the line and not the claim. It ran the summarized tests, and three findings rest on it and not on the law: its 64-bit integers hold pairs to  $2^{63}$ , below nature's den of order  $2^{94}$  for the electron (S.54), the law fixing no width, and its Link unit is 16, two bounds that are not physics; the closed box's swing, the potentials' swing growing linearly over the declared run, a lattice reading of the archive's runs, its one named cure the law's one-way face, not built (the long version's Section 6.5 and S.62 at its tag); and the ion's telegraph, a run not made (S.50 (15)). **Author contributions.** The author conceived the model, carried out the derivations and the computations with the tool stated below, and wrote the paper. **The use of a large language model.** The author used a large language model (Claude, Anthropic, through Claude Code, 2026) as a tool under the author's direction for the software, the computations, the derivations and their checks and the drafting of the text, including the cut of this version from the longer one in the archive; the model is not an author, and the author reviewed every statement and takes full responsibility for the content.

## References

- [1] P. Ehrenfest, Adiabatische Invarianten und Quantentheorie, *Annalen der Physik* 356, 327 (1916), doi:10.1002/andp.19163561905.
- [2] K. Zuse, *Rechnender Raum* (Vieweg, Braunschweig, 1969); English translation: *Calculating Space*, MIT Technical Translation AZT-70-164-GEMIT (MIT Project MAC, Cambridge, 1970).
- [3] T. Toffoli and N. Margolus, Invertible cellular automata: a review, *Physica D* 45, 229 (1990), doi:10.1016/0167-2789(90)90185-R.
- [4] D. J. D. Earn and S. Tremaine, Exact numerical studies of Hamiltonian maps: iterating without roundoff error, *Physica D* 56, 1 (1992), doi:10.1016/0167-2789(92)90047-Q.
- [5] H.-T. Elze, Action principle for cellular automata and the linearity of quantum mechanics, *Physical Review A* 89, 012111 (2014), doi:10.1103/PhysRevA.89.012111.
- [6] I. Bialynicki-Birula, Weyl, Dirac, and Maxwell equations on a lattice as unitary cellular automata, *Physical Review D* 49, 6920 (1994), doi:10.1103/PhysRevD.49.6920.
- [7] S. Wolfram, *A New Kind of Science* (Wolfram Media, Champaign, 2002).
- [8] E. Fredkin, Digital mechanics: an informational process based on reversible universal cellular automata, *Physica D* 45, 254 (1990), doi:10.1016/0167-2789(90)90186-S.

- [9] P. Arrighi, An overview of quantum cellular automata, *Natural Computing* 18, 885 (2019), doi:10.1007/s11047-019-09762-6.
- [10] H. De Raedt, K. De Raedt and K. Michielsen, Event-based simulation of single-photon beam splitters and Mach-Zehnder interferometers, *Europhysics Letters* 69, 861 (2005), doi:10.1209/epl/i2004-10443-7.
- [11] G. 't Hooft, *The Cellular Automaton Interpretation of Quantum Mechanics, Fundamental Theories of Physics* 185 (Springer, Cham, 2016), doi:10.1007/978-3-319-41285-6.
- [12] J. A. Wheeler and R. P. Feynman, Interaction with the absorber as the mechanism of radiation, *Reviews of Modern Physics* 17, 157 (1945), doi:10.1103/RevModPhys.17.157.
- [13] J. G. Cramer, The transactional interpretation of quantum mechanics, *Reviews of Modern Physics* 58, 647 (1986), doi:10.1103/RevModPhys.58.647.
- [14] Y. Aharonov and L. Vaidman, The two-state vector formalism: an updated review, *Lecture Notes in Physics* 734, 399 (Springer, Berlin, 2008), doi:10.1007/978-3-540-73473-4.13.
- [15] K. B. Wharton and N. Argaman, Colloquium: Bell's theorem and locally mediated reformulations of quantum mechanics, *Reviews of Modern Physics* 92, 021002 (2020), doi:10.1103/RevModPhys.92.021002.
- [16] H. Yilmaz, New approach to general relativity, *Physical Review* 111, 1417 (1958), doi:10.1103/PhysRev.111.1417.
- [17] R. H. Dicke, Gravitation without a principle of equivalence, *Reviews of Modern Physics* 29, 363 (1957), doi:10.1103/RevModPhys.29.363.
- [18] H. E. Puthoff, Polarizable-vacuum (PV) approach to general relativity, *Foundations of Physics* 32, 927 (2002), doi:10.1023/A:1016011413407.
- [19] P. Boonserm, T. Ngampitipan, A. Simpson and M. Visser, Exponential metric represents a traversable wormhole, *Physical Review D* 98, 084048 (2018), doi:10.1103/PhysRevD.98.084048.
- [20] K. G. Wilson, Confinement of quarks, *Physical Review D* 10, 2445 (1974), doi:10.1103/PhysRevD.10.2445.
- [21] A. Sciarretta, A local-realistic model of quantum mechanics based on a discrete spacetime, *Foundations of Physics* 48, 60 (2018), doi:10.1007/s10701-017-0129-9.
- [22] D. Bohm, A suggested interpretation of the quantum theory in terms of "hidden" variables. I, *Physical Review* 85, 166 (1952), doi:10.1103/PhysRev.85.166.

- [23] C. Philippidis, C. Dewdney and B. J. Hiley, Quantum interference and the quantum potential, *Il Nuovo Cimento B* 52, 15 (1979), doi:10.1007/BF02743566.
- [24] R. P. Feynman, Simulating physics with computers, *International Journal of Theoretical Physics* 21, 467 (1982), doi:10.1007/BF02650179.
- [25] J. Collins, A. Perez, D. Sudarsky, L. Urrutia and H. Vucetich, Lorentz invariance and quantum gravity: an additional fine-tuning problem?, *Physical Review Letters* 93, 191301 (2004), doi:10.1103/PhysRevLett.93.191301.
- [26] D. Mattingly, Modern tests of Lorentz invariance, *Living Reviews in Relativity* 8, 5 (2005), doi:10.12942/lrr-2005-5.
- [27] A. Gonen, Universe24: an integer cellular automaton read through clicks, the formulas it shares with nature and the ones it adds, the long version of this article at the archive's tag `paper-long-v1.1` (2026), <https://github.com/Closer24/Universe24/tree/paper-long-v1.1>.
- [28] The law of the lattice, `docs/ALGEBRA.md` of the archive [94] at the tagged release 1.1.0 (the release's commit; the implementation is at the commit 1fe3790a of 2026-10-05, which the documents name), cited by the name of its section: the lattice and the families (the objects, a family's declaration, the interval, readings and measurements); Rule3 (the line, the direction, the four acts, the paces, the bound, the self-source, the conserved form); the click (the booking, the count is the record's share, the click is the detector's report, no click names a Node, the algebra of the exclusivity); the primitives; the stable body (what a body is, the well, the functional and the dilation, the surplus leaves, the generator, the velocity); what the law says of nature (the reading, with no waves on the event lattice, what produces a click is not known and nature's numbers enter the lattice only as clicks; the hypotheses under their own names; the postulates; the constants; the rows against nature); what is open.
- [29] The decisions, `docs/HIGHLIGHTS.md` of the archive [94] at the tagged release 1.1.0: the model's design decisions, one line each, cited by the line's name.
- [30] P. Arrighi and S. Facchini, Decoupled quantum walks, models of the Klein-Gordon and wave equations, *EPL (Europhysics Letters)* 104, 60004 (2013), doi:10.1209/0295-5075/104/60004.
- [31] T. E. Hull and A. R. Dobell, Random number generators, *SIAM Review* 4, 230 (1962), doi:10.1137/1004061.
- [32] M. Born, Zur Quantenmechanik der Stoßvorgänge, *Zeitschrift für Physik* 37, 863 (1926), doi:10.1007/BF01397477.
- [33] P. Grangier, G. Roger and A. Aspect, Experimental evidence for a photon anti-correlation effect on a beam splitter: a new light on single-photon interferences,

- Europhysics Letters 1, 173 (1986), doi:10.1209/0295-5075/1/4/004.
- [34] A. H. Compton and A. W. Simon, Directed quanta of scattered X-rays, *Physical Review* 26, 289 (1925), doi:10.1103/PhysRev.26.289.
  - [35] A. H. Compton, A quantum theory of the scattering of X-rays by light elements, *Physical Review* 21, 483 (1923), doi:10.1103/PhysRev.21.483.
  - [36] H. Yukawa, On the interaction of elementary particles. I, *Proceedings of the Physico-Mathematical Society of Japan* 17, 48 (1935), doi:10.11429/ppmsj1919.17.0\_48.
  - [37] B. P. Abbott et al. (LIGO Scientific Collaboration, Virgo Collaboration, Fermi Gamma-ray Burst Monitor and INTEGRAL), Gravitational waves and gamma-rays from a binary neutron star merger: GW170817 and GRB 170817A, *The Astrophysical Journal Letters* 848, L13 (2017), doi:10.3847/2041-8213/aa920c.
  - [38] G. H. Derrick, Comments on nonlinear wave equations as models for elementary particles, *Journal of Mathematical Physics* 5, 1252 (1964), doi:10.1063/1.1704233.
  - [39] S. I. Pohozaev, On the eigenfunctions of the equation  $\Delta u + \lambda f(u) = 0$ , *Doklady Akademii Nauk SSSR* 165, 36 (1965).
  - [40] E. A. Milne, A Newtonian expanding universe, *The Quarterly Journal of Mathematics os-5*, 64 (1934), doi:10.1093/qmath/os-5.1.64.
  - [41] W. H. McCrea and E. A. Milne, Newtonian universes and the curvature of space, *The Quarterly Journal of Mathematics os-5*, 73 (1934), doi:10.1093/qmath/os-5.1.73.
  - [42] P. J. Mohr, D. B. Newell, B. N. Taylor and E. Tiesinga, CODATA recommended values of the fundamental physical constants: 2022, *Reviews of Modern Physics* 97, 025002 (2025), doi:10.1103/RevModPhys.97.025002.
  - [43] Z. Cao et al. (LHAASO Collaboration), Ultrahigh-energy photons up to 1.4 petaelectronvolts from 12 gamma-ray Galactic sources, *Nature* 594, 33 (2021), doi:10.1038/s41586-021-03498-z.
  - [44] V. A. Acciari et al. (MAGIC Collaboration), Bounds on Lorentz invariance violation from MAGIC observation of GRB 190114C, *Physical Review Letters* 125, 021301 (2020), doi:10.1103/PhysRevLett.125.021301.
  - [45] R. V. Pound and G. A. Rebka, Apparent weight of photons, *Physical Review Letters* 4, 337 (1960), doi:10.1103/PhysRevLett.4.337.
  - [46] C. M. Will, The confrontation between general relativity and experiment, *Living Reviews in Relativity* 17, 4 (2014), doi:10.12942/lrr-2014-4.

- [47] B. Bertotti, L. Iess and P. Tortora, A test of general relativity using radio links with the Cassini spacecraft, *Nature* 425, 374 (2003), doi:10.1038/nature01997.
- [48] I. I. Shapiro, Fourth test of general relativity, *Physical Review Letters* 13, 789 (1964), doi:10.1103/PhysRevLett.13.789.
- [49] C. W. F. Everitt et al., Gravity Probe B: final results of a space experiment to test general relativity, *Physical Review Letters* 106, 221101 (2011), doi:10.1103/PhysRevLett.106.221101.
- [50] L. Shao and N. Wex, New tests of local Lorentz invariance of gravity with small-eccentricity binary pulsars, *Classical and Quantum Gravity* 29, 215018 (2012), doi:10.1088/0264-9381/29/21/215018.
- [51] K. Nordtvedt, Probing gravity to the second post-Newtonian order and to one part in  $10^7$  using the spin axis of the Sun, *The Astrophysical Journal* 320, 871 (1987), doi:10.1086/165603.
- [52] L. Shao, R. N. Caballero, M. Kramer, N. Wex, D. J. Champion and A. Jessner, A new limit on local Lorentz invariance violation of gravity from solitary pulsars, *Classical and Quantum Gravity* 30, 165019 (2013), doi:10.1088/0264-9381/30/16/165019.
- [53] B. Altschul, Lorentz violation and synchrotron radiation, *Physical Review D* 72, 085003 (2005), doi:10.1103/PhysRevD.72.085003.
- [54] F. W. Stecker, Limiting superluminal electron and neutrino velocities using the 2010 Crab Nebula flare and the IceCube PeV neutrino events, *Astroparticle Physics* 56, 16 (2014), doi:10.1016/j.astropartphys.2014.02.007.
- [55] S. Coleman and S. L. Glashow, High-energy tests of Lorentz invariance, *Physical Review D* 59, 116008 (1999), doi:10.1103/PhysRevD.59.116008.
- [56] P. Touboul et al. (MICROSCOPE Collaboration), MICROSCOPE mission: final results of the test of the equivalence principle, *Physical Review Letters* 129, 121102 (2022), doi:10.1103/PhysRevLett.129.121102.
- [57] G. Bressi, G. Carugno, F. Della Valle, G. Galeazzi, G. Ruoso and G. Sartori, Testing the neutrality of matter by acoustic means in a spherical resonator, *Physical Review A* 83, 052101 (2011), doi:10.1103/PhysRevA.83.052101.
- [58] J.-W. Pan, D. Bouwmeester, M. Daniell, H. Weinfurter and A. Zeilinger, Experimental test of quantum nonlocality in three-photon Greenberger-Horne-Zeilinger entanglement, *Nature* 403, 515 (2000), doi:10.1038/35000514.
- [59] The Event Horizon Telescope Collaboration, First Sagittarius A\* Event Horizon Telescope results. VI. Testing the black hole metric, *The Astrophysical Journal*

Letters 930, L17 (2022), doi:10.3847/2041-8213/ac6756.

- [60] D. Psaltis et al. (The Event Horizon Telescope Collaboration), Gravitational test beyond the first post-Newtonian order with the shadow of the M87 black hole, *Physical Review Letters* 125, 141104 (2020), doi:10.1103/PhysRevLett.125.141104.
- [61] R. C. Tolman, P. Ehrenfest and B. Podolsky, On the gravitational field produced by light, *Physical Review* 37, 602 (1931), doi:10.1103/PhysRev.37.602.
- [62] M. Zych, F. Costa, I. Pikovski and C. Brukner, Quantum interferometric visibility as a witness of general relativistic proper time, *Nature Communications* 2, 505 (2011), doi:10.1038/ncomms1498.
- [63] A. D. V. Di Virgilio, J. Belfi, W.-T. Ni, N. Beverini, G. Carelli, E. Maccioni and A. Porzio, GINGER: a feasibility study, *The European Physical Journal Plus* 132, 157 (2017), doi:10.1140/epjp/i2017-11452-6.
- [64] B. S. Cirel'son, Quantum generalizations of Bell's inequality, *Letters in Mathematical Physics* 4, 93 (1980), doi:10.1007/BF00417500.
- [65] J. S. Bell, On the Einstein Podolsky Rosen paradox, *Physics Physique Fizika* 1, 195 (1964), doi:10.1103/PhysicsPhysiqueFizika.1.195.
- [66] J. F. Clauser, M. A. Horne, A. Shimony and R. A. Holt, Proposed experiment to test local hidden-variable theories, *Physical Review Letters* 23, 880 (1969), doi:10.1103/PhysRevLett.23.880.
- [67] P. M. Pearle, Hidden-variable example based upon data rejection, *Physical Review D* 2, 1418 (1970), doi:10.1103/PhysRevD.2.1418.
- [68] P. H. Eberhard, Background level and counter efficiencies required for a loophole-free Einstein-Podolsky-Rosen experiment, *Physical Review A* 47, R747 (1993), doi:10.1103/PhysRevA.47.R747.
- [69] A. Garg and N. D. Mermin, Detector inefficiencies in the Einstein-Podolsky-Rosen experiment, *Physical Review D* 35, 3831 (1987), doi:10.1103/PhysRevD.35.3831.
- [70] H. De Raedt, K. De Raedt, K. Michielsen, K. Keimpema and S. Miyashita, Event-based computer simulation model of Aspect-type experiments strictly satisfying Einstein's locality conditions, *Journal of the Physical Society of Japan* 76, 104005 (2007), doi:10.1143/JPSJ.76.104005.
- [71] K. Michielsen and H. De Raedt, Event-based simulation of quantum physics experiments, *International Journal of Modern Physics C* 25, 1430003 (2014), doi:10.1142/S0129183114300036.

- [72] B. Hensen et al., Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres, *Nature* 526, 682 (2015), doi:10.1038/nature15759.
- [73] M. Giustina et al., Significant-loophole-free test of Bell’s theorem with entangled photons, *Physical Review Letters* 115, 250401 (2015), doi:10.1103/PhysRevLett.115.250401.
- [74] L. K. Shalm et al., Strong loophole-free test of local realism, *Physical Review Letters* 115, 250402 (2015), doi:10.1103/PhysRevLett.115.250402.
- [75] D. M. Greenberger, M. A. Horne and A. Zeilinger, Going beyond Bell’s theorem, in *Bell’s Theorem, Quantum Theory and Conceptions of the Universe*, ed. M. Kafatos (Kluwer, Dordrecht, 1989), 69, doi:10.1007/978-94-017-0849-4\_10.
- [76] N. D. Mermin, Extreme quantum entanglement in a superposition of macroscopically distinct states, *Physical Review Letters* 65, 1838 (1990), doi:10.1103/PhysRevLett.65.1838.
- [77] S. Hossenfelder and T. Palmer, Rethinking superdeterminism, *Frontiers in Physics* 8, 139 (2020), doi:10.3389/fphy.2020.00139.
- [78] B. P. Abbott et al. (LIGO Scientific Collaboration and Virgo Collaboration), GW170814: a three-detector observation of gravitational waves from a binary black hole coalescence, *Physical Review Letters* 119, 141101 (2017), doi:10.1103/PhysRevLett.119.141101.
- [79] A. G. Riess et al., Observational evidence from supernovae for an accelerating universe and a cosmological constant, *The Astronomical Journal* 116, 1009 (1998), doi:10.1086/300499.
- [80] S. Perlmutter et al. (The Supernova Cosmology Project), Measurements of  $\Omega$  and  $\Lambda$  from 42 high-redshift supernovae, *The Astrophysical Journal* 517, 565 (1999), doi:10.1086/307221.
- [81] D. Brout et al., The Pantheon+ analysis: cosmological constraints, *The Astrophysical Journal* 938, 110 (2022), doi:10.3847/1538-4357/ac8e04.
- [82] S. Navas et al. (Particle Data Group), Review of particle physics, *Physical Review D* 110, 030001 (2024), doi:10.1103/PhysRevD.110.030001.
- [83] F. J. Hasert et al., Observation of neutrino-like interactions without muon or electron in the Gargamelle neutrino experiment, *Physics Letters B* 46, 138 (1973), doi:10.1016/0370-2693(73)90499-1.
- [84] The ALEPH, DELPHI, L3, OPAL and SLD Collaborations, Precision electroweak measurements on the Z resonance, *Physics Reports* 427, 257 (2006), doi:10.1016/j.physrep.2005.12.006.

- [85] C. S. Wu, E. Ambler, R. W. Hayward, D. D. Hoppes and R. P. Hudson, Experimental test of parity conservation in beta decay, *Physical Review* 105, 1413 (1957), doi:10.1103/PhysRev.105.1413.
- [86] A. Sommerfeld, Zur Quantentheorie der Spektrallinien, *Annalen der Physik* 356, 1 (1916), doi:10.1002/andp.19163561702.
- [87] C. C. J. Roothaan, L. M. Sachs and A. W. Weiss, Analytical self-consistent field functions for the atomic configurations  $1s^2$ ,  $1s^22s$  and  $1s^22s^2$ , *Reviews of Modern Physics* 32, 186 (1960), doi:10.1103/RevModPhys.32.186.
- [88] M. A. Rowe, D. Kielpinski, V. Meyer, C. A. Sackett, W. M. Itano, C. Monroe and D. J. Wineland, Experimental violation of a Bell's inequality with efficient detection, *Nature* 409, 791 (2001), doi:10.1038/35057215.
- [89] S. Storz et al., Loophole-free Bell inequality violation with superconducting circuits, *Nature* 617, 265 (2023), doi:10.1038/s41586-023-05885-0.
- [90] V. Vasileiou et al., Constraints on Lorentz invariance violation from Fermi-Large Area Telescope observations of gamma-ray bursts, *Physical Review D* 87, 122001 (2013), doi:10.1103/PhysRevD.87.122001.
- [91] R. L. Mössbauer, Kernresonanzfluoreszenz von Gammastrahlung in  $\text{Ir}^{191}$ , *Zeitschrift für Physik* 151, 124 (1958), doi:10.1007/BF01344210.
- [92] R. J. Cook and H. J. Kimble, Possibility of direct observation of quantum jumps, *Physical Review Letters* 54, 1023 (1985), doi:10.1103/PhysRevLett.54.1023.
- [93] J. C. Bergquist, R. G. Hulet, W. M. Itano and D. J. Wineland, Observation of quantum jumps in a single atom, *Physical Review Letters* 57, 1699 (1986), doi:10.1103/PhysRevLett.57.1699.
- [94] A. Gonen, Universe24: the implementation of one integer rule on a cubic lattice, read through clicks, Zenodo (2026), the concept DOI, the record of the version at the tag `paper-long-v1.1` to be deposited before submission, doi:10.5281/zenodo.22738745.